CHAPTER 02

# Isomers

## Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 3790c3072a422920a2793636cc0e671c8e2a766731209b66b68c3e110873ccb4

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

### Coverage checklist

ch02:exercise:isomer-pair | isomer-pair:stage-1, isomer-pair:stage-2, isomer-pair:stage-3

ch02:exercise:butane-rotation | butane-rotation:stage-1, butane-rotation:stage-2, butane-rotation:stage-3

ch02:exercise:stereocenters | stereocenters:stage-1, stereocenters:stage-2, stereocenters:stage-3

ch02:exercise:fischer-cip | fischer-cip:stage-1, fischer-cip:stage-2, fischer-cip:stage-3

ch02:exercise:ez-ee | ez-ee:stage-1, ez-ee:stage-2, ez-ee:stage-3

ch02:lesson:connectivity | connectivity:rule, connectivity:cue, connectivity:trap, connectivity:app, connectivity:step-1, connectivity:step-2, connectivity:step-3

ch02:lesson:conformations | conformations:rule, conformations:cue, conformations:trap, conformations:app, conformations:step-1, conformations:step-2

ch02:lesson:chirality | chirality:rule, chirality:cue, chirality:trap, chirality:app, chirality:step-1, chirality:step-2

ch02:lesson:configuration | configuration:rule, configuration:cue, configuration:trap, configuration:app, configuration:step-1, configuration:step-2, configuration:step-3

ch02:lesson:alkene | alkene:rule, alkene:cue, alkene:trap, alkene:app, alkene:step-1, alkene:step-2

ch02:molecule:ethanol | ethanol

ch02:molecule:ether | ether

ch02:molecule:isobutane | isobutane

ch02:molecule:butane | butane

ch02:molecule:cyclohexane | cyclohexane

ch02:molecule:butanol2 | butanol2

ch02:molecule:butanediol | butanediol

ch02:molecule:e-butene | e-butene

ch02:molecule:z-butene | z-butene

ch02:curriculum | scope-1, scope-2, scope-3

# Curriculum scope

scope-1: 2.1 Structural Isomers

scope-2: 2.2 Stereoisomers

scope-3: 2.3 Relative and Absolute Configurations

Objective: Classify isomer relationships

Objective: Assign CIP and E/Z descriptors

Objective: Analyze conformations, symmetry, and optical mixtures

Supplied Kaplan curriculum: Chapter 2; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter02b.xhtml

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

# connectivity | Connectivity vs. spatial arrangement

Same formula is only the first test

## connectivity:rule | Core Rules

Constitutional isomers share a molecular formula but differ in connectivity. Stereoisomers have the same connectivity and differ in spatial arrangement. Conformers interconvert by rotations about single bonds.

## connectivity:cue | Memory Cues

Formula → connections → space. Compare in that order.

## connectivity:trap | Common Mistakes / MCAT Traps

A different drawing orientation does not prove isomerism. Rotating the whole molecule changes neither connectivity nor configuration.

## connectivity:app | MCAT Applications

An alcohol and an ether can share a formula but differ in hydrogen-bond donation, boiling point, and reactivity.

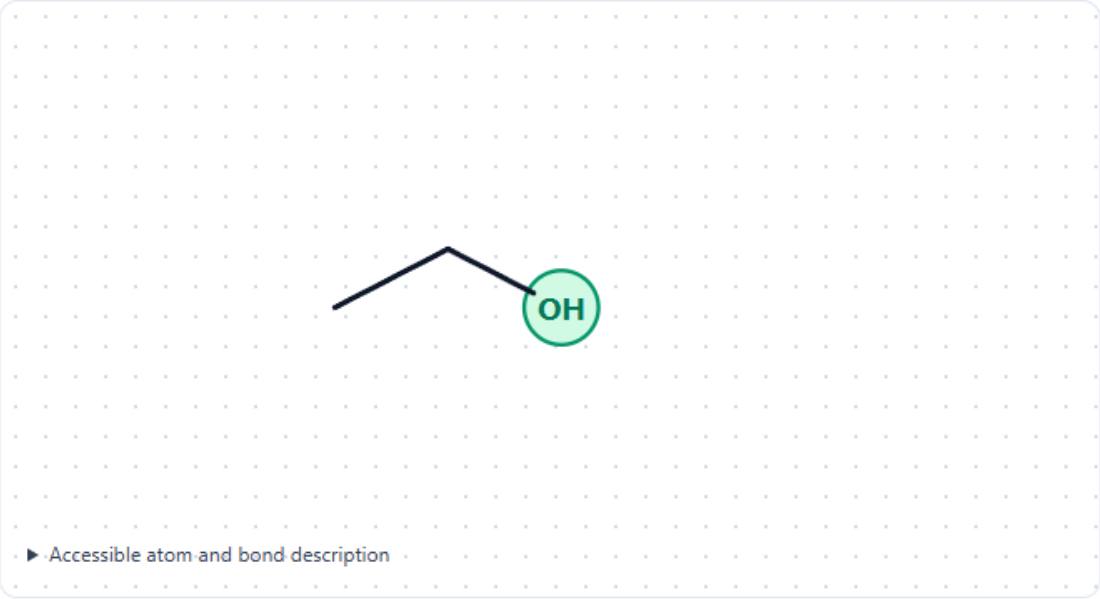
## connectivity:step-1 | 1. Alcohol connectivity

Count two carbons, six hydrogens, and one oxygen. The oxygen bears H and can donate a hydrogen bond.

Displayed molecule ID: ethanol

{"highlights": {"prin": [2]}, "revealName": true}

Screenshot: connectivity-step-1



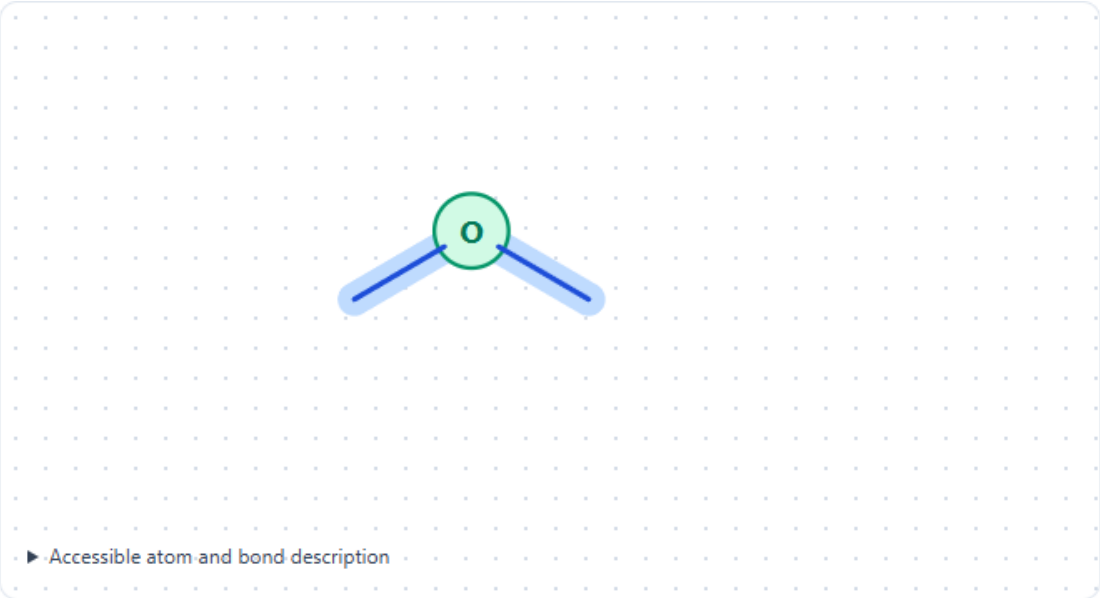

## connectivity:step-2 | 2. Ether connectivity

The same formula now has oxygen between two carbons. No O–H bond is present. This is a constitutional isomer, not a different conformation.

Displayed molecule ID: ether

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Screenshot: connectivity-step-2



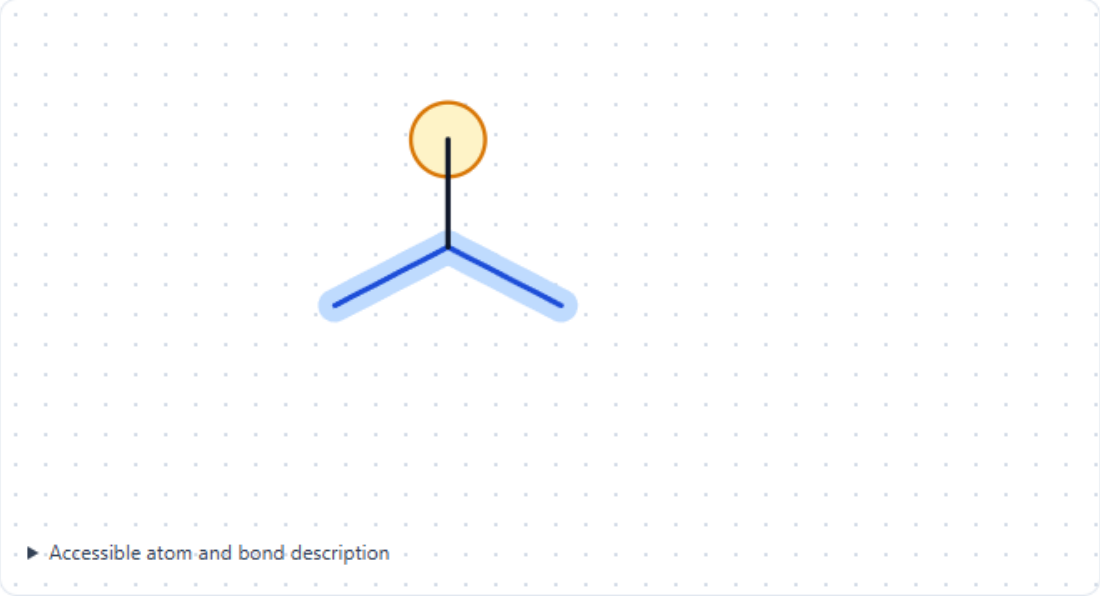

## connectivity:step-3 | 3. Branched framework

Butane and 2-methylpropane are another constitutional-isomer pair: identical formula, different carbon skeletons.

Displayed molecule ID: isobutane

{"highlights": {"path": [0, 1, 2], "sub": [3]}, "revealName": true}

Screenshot: connectivity-step-3



# conformations | Conformations and ring strain

Rotation changes energy, not identity

## conformations:rule | Core Rules

In butane, anti and gauche are staggered conformations; eclipsed conformations have greater torsional strain. Cyclohexane chairs reduce angle and torsional strain. A ring flip exchanges axial and equatorial positions while retaining up/down orientation.

## conformations:cue | Memory Cues

Anti = 180° between the methyl groups. Gauche = 60°. In a chair, bulky groups usually prefer equatorial sites.

## conformations:trap | Common Mistakes / MCAT Traps

Axial is not synonymous with up: axial directions alternate around the ring. A chair flip preserves cis/trans relationships.

## conformations:app | MCAT Applications

Conformational preferences affect how a flexible drug fits a binding site; a more populated conformer need not be the bound conformer.

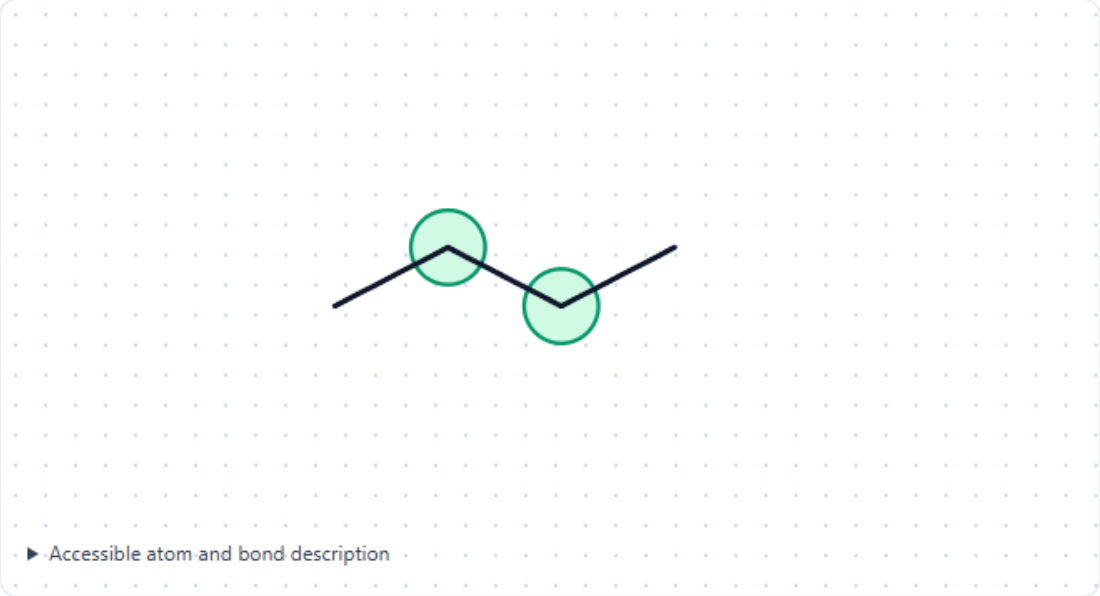
## conformations:step-1 | 1. Rotate a single bond

Look down the middle C–C bond of butane. Adjust the angle to compare torsional and methyl–methyl interactions.

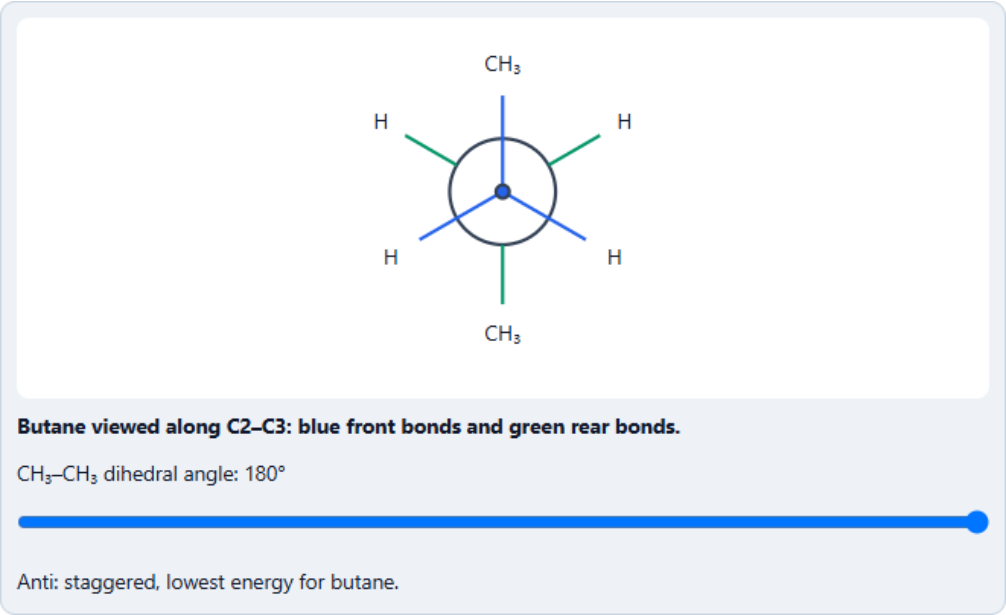
Displayed molecule ID: butane

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: conformations-step-1



Screenshot: conformations-step-1-diagram




{"diagramSpecification": {"type": "newman", "caption": "Butane viewed along C2–C3: blue front bonds and green rear bonds."}}

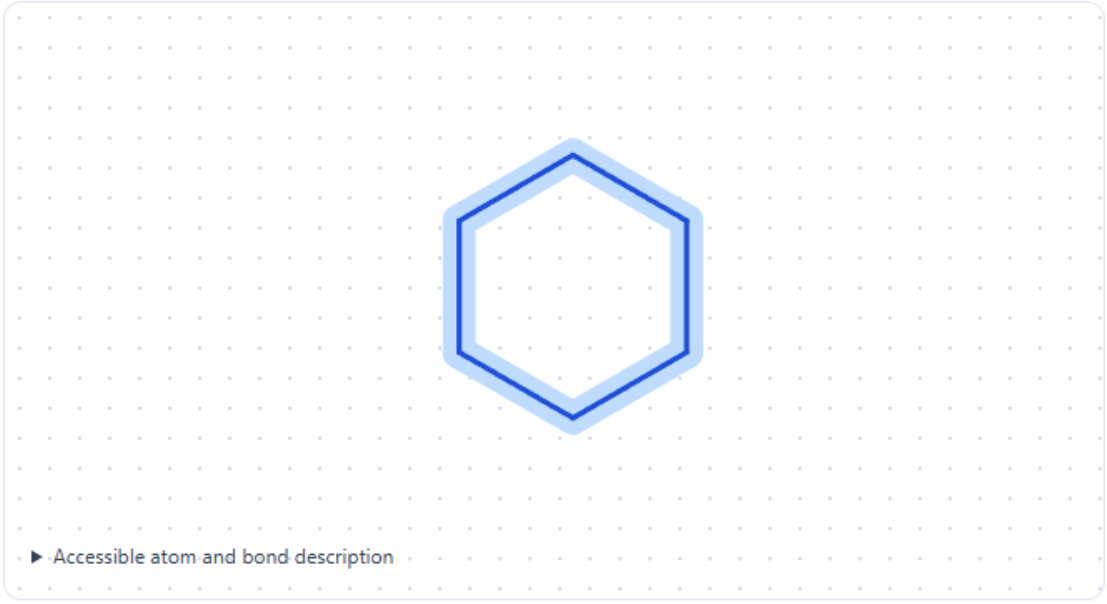
## conformations:step-2 | 2. Ring constraints

This hexagon shows connectivity only, not a flat equilibrium geometry. Cyclohexane normally adopts a chair. An axial substituent interacts with axial groups at C3 and C5; an equatorial position avoids much of this crowding.

Displayed molecule ID: cyclohexane

{"highlights": {"path": [0, 1, 2, 3, 4, 5]}, "revealName": true}

Screenshot: conformations-step-2



# chirality | Chirality, enantiomers, and meso compounds

Mirror image ≠ always a different molecule

## chirality:rule | Core Rules

A tetrahedral carbon with four different substituents is a common stereogenic center. Enantiomers are nonsuperimposable mirror images. Diastereomers are stereoisomers that are not mirror images; E/Z pairs qualify too.

## chirality:cue | Memory Cues

One inverted center in a one-center molecule gives its enantiomer. With several centers, compare the entire molecule.

## chirality:trap | Common Mistakes / MCAT Traps

$2^n$ is a maximum number of stereoisomers, not a guarantee. Internal symmetry can make a molecule with stereocenters achiral (meso). R/S does not predict the sign of optical rotation.

## chirality:app | MCAT Applications

Enantiomers have equal properties in achiral environments except opposite optical rotations; chiral receptors can distinguish them.

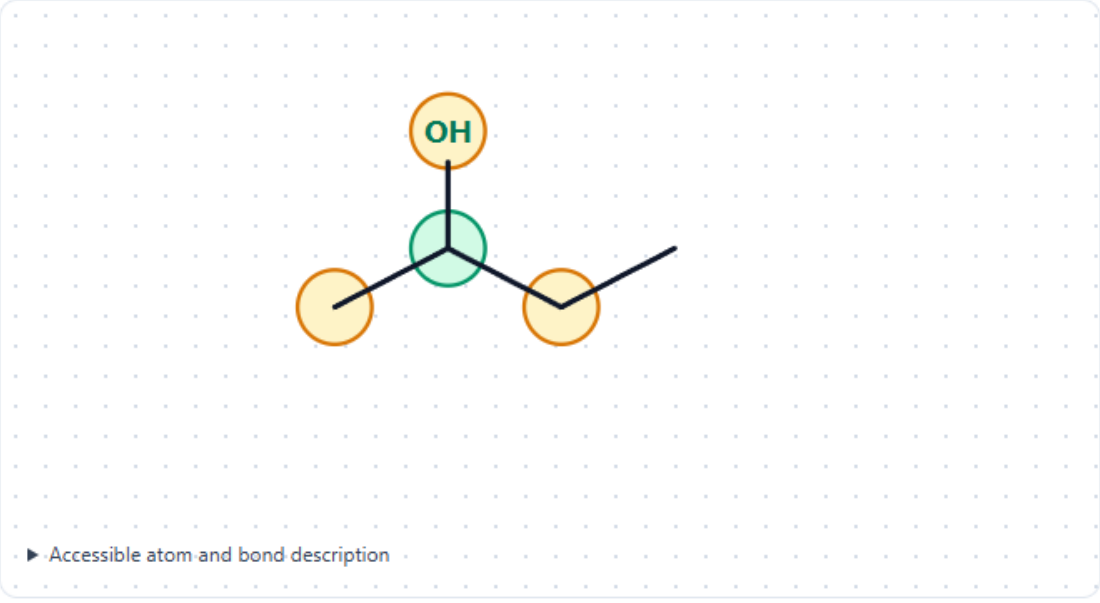
## chirality:step-1 | 1. Identify four groups

The alcohol-bearing carbon is bonded to OH, H, methyl, and ethyl: four different groups. The flat connectivity drawing alone does not assign R or S.

Displayed molecule ID: butanol2

{"highlights": {"prin": [1], "sub": [0, 2, 4]}, "revealName": false}

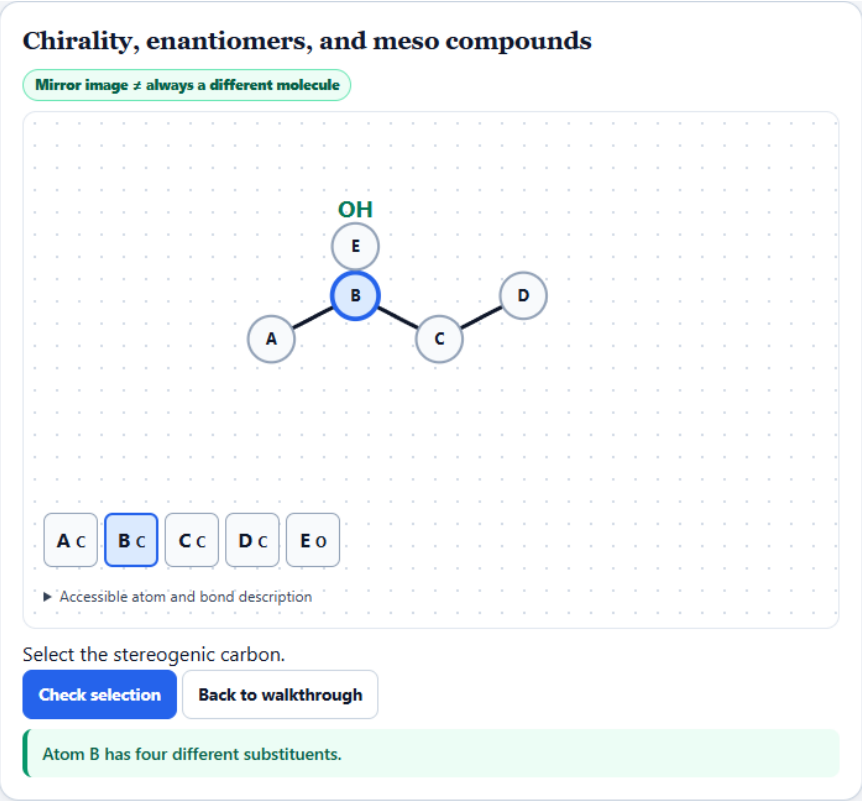
Screenshot: chirality-step-1




Embedded challenge: Select the stereogenic carbon.

{"prompt": "Select the stereogenic carbon.", "answer": [1], "explanation": "Atom B has four different substituents."}

Screenshot: chirality-step-1-challenge



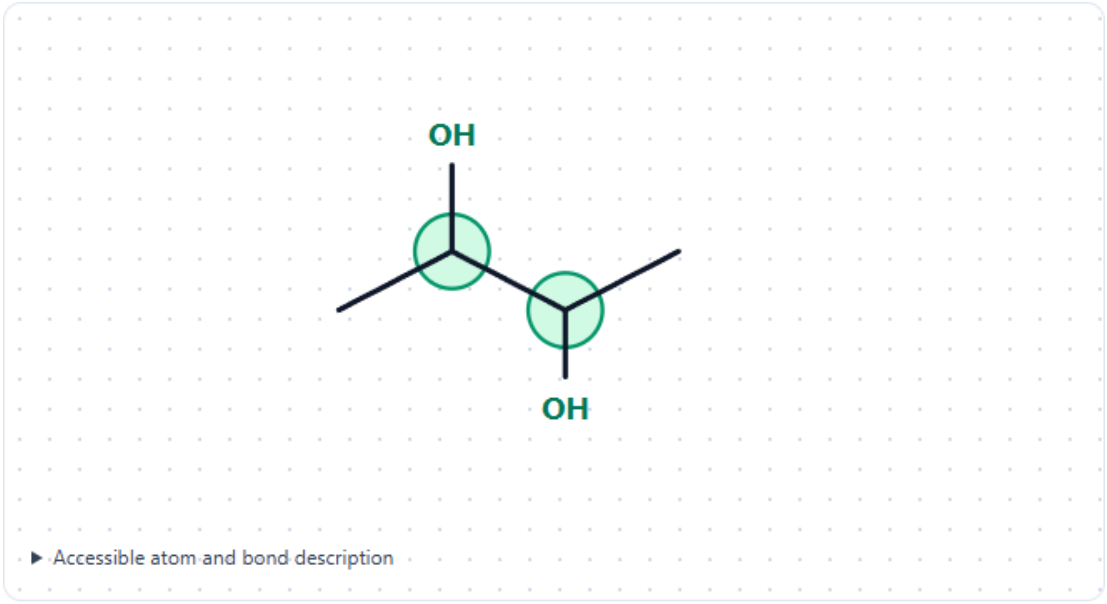

## chirality:step-2 | 2. Count with symmetry

Butane-2,3-diol has two potential stereocenters. Its (2R,3S) and (2S,3R) descriptions refer to the same meso compound because of symmetry; the RR and SS forms are an enantiomeric pair. Total: three stereoisomers.

Displayed molecule ID: butanediol

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: chirality-step-2



# configuration | CIP priorities and Fischer projections

Assign priority before tracing direction

## configuration:rule | Core Rules

Rank directly attached atoms by atomic number; isotopes by mass. Resolve ties one sphere at a time using sorted atom lists. Treat multiple bonds with duplicate-atom bookkeeping. With priority 4 away, clockwise 1→2→3 is R and counterclockwise is S.

## configuration:cue | Memory Cues

Fischer: horizontal comes forward; vertical goes back. If group 4 is horizontal, reverse the apparent result.

## configuration:trap | Common Mistakes / MCAT Traps

A 180° in-plane Fischer rotation preserves configuration; a 90° rotation does not. D/L is a relative-configuration convention, not a synonym for R/S or +/−.

## configuration:app | MCAT Applications

Most proteinogenic L amino acids are S; L cysteine is R because sulfur changes CIP ranking. Glycine has no stereogenic alpha carbon.

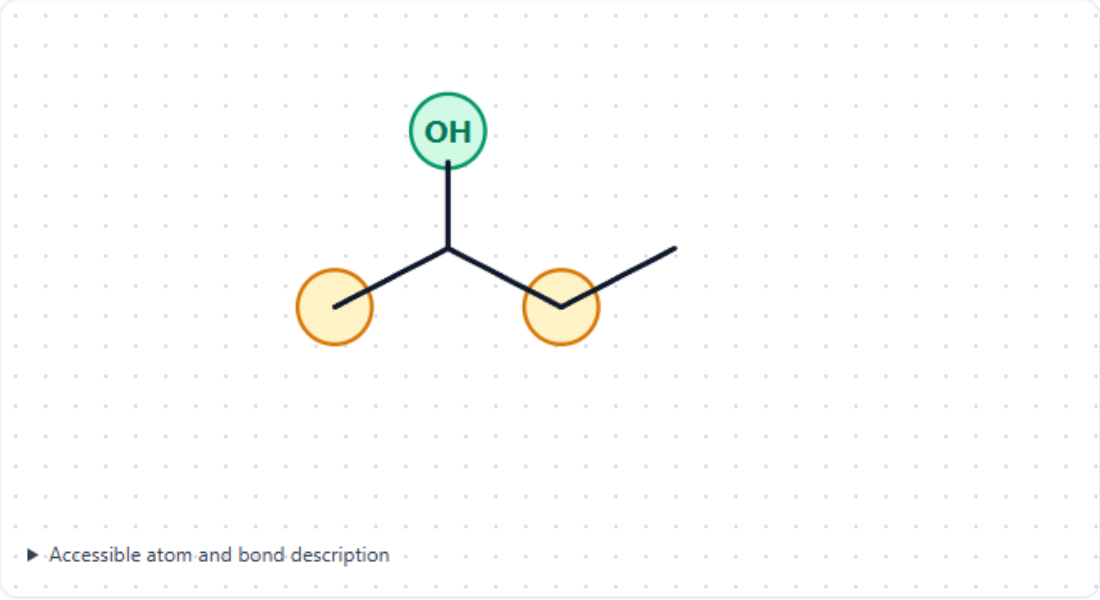
## configuration:step-1 | 1. Rank the substituents

OH ranks first because O outranks C. Ethyl outranks methyl by the next sphere: [C,H,H] beats [H,H,H]. H is fourth.

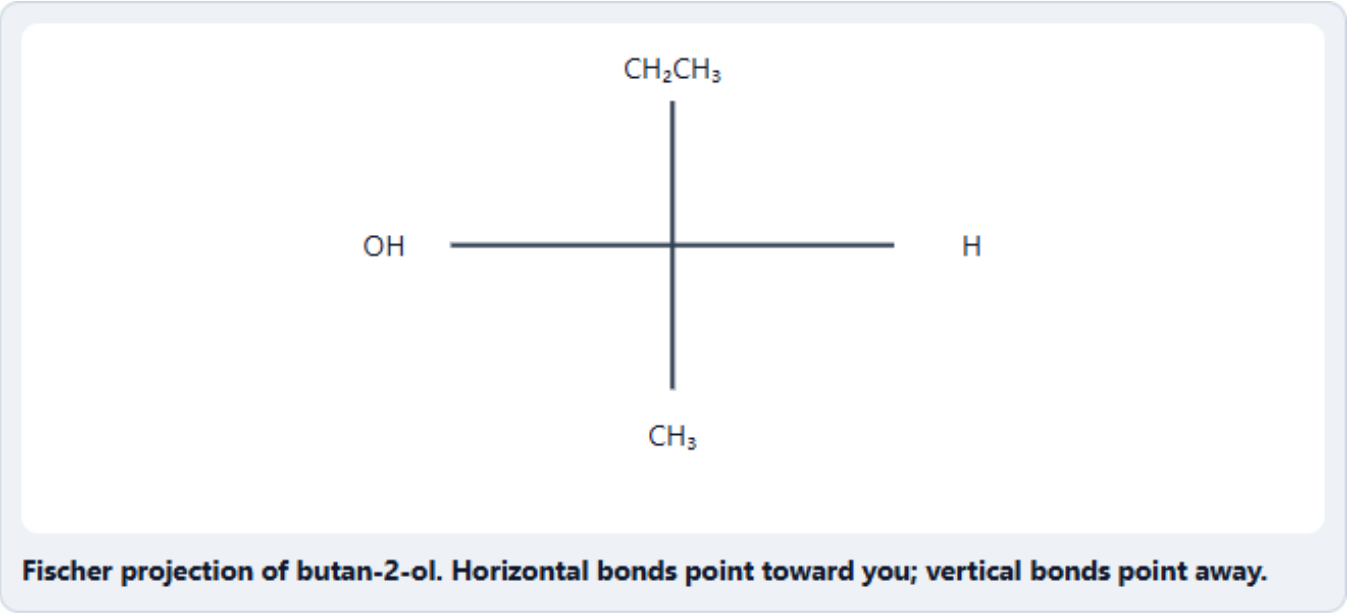
Displayed molecule ID: butanol2

{"highlights": {"prin": [4], "sub": [2, 0]}, "revealName": false}

Screenshot: configuration-step-1



Screenshot: configuration-step-1-diagram




**Fischer projection of butan-2-ol. Horizontal bonds point toward you; vertical bonds point away.**

{"diagramSpecification": {"type": "fischer", "top": "CH□CH□", "bottom": "CH□", "left": "OH", "right": "H", "caption": "Fischer projection of butan-2-ol. Horizontal bonds point toward you; vertical bonds point away."}}

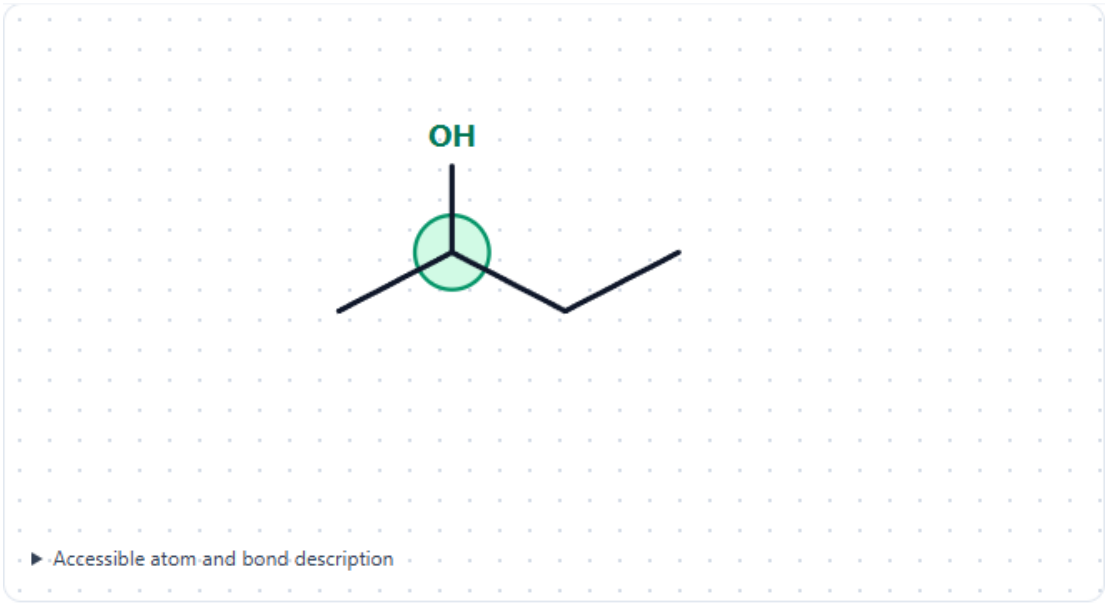
## configuration:step-2 | 2. Orient then assign

In this Fischer projection, 1→2→3 appears clockwise, but H is horizontal and points toward you. Reverse the apparent sense: the projection is S.

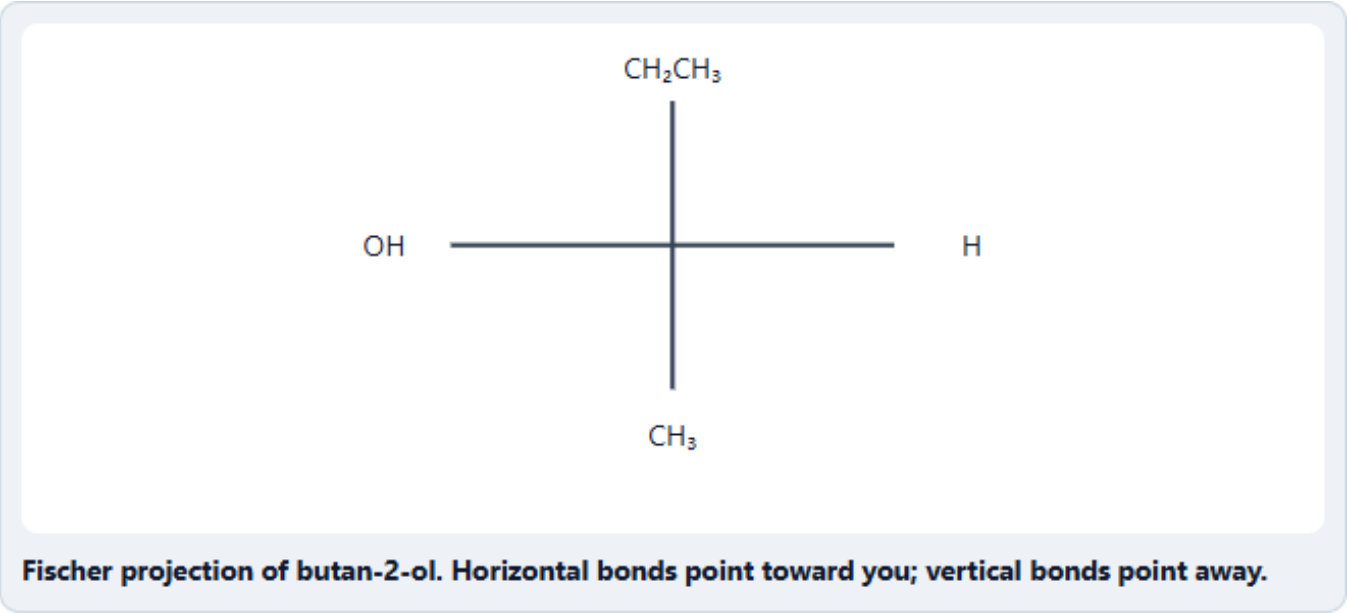
Displayed molecule ID: butanol2

{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: configuration-step-2



Screenshot: configuration-step-2-diagram




**Fischer projection of butan-2-ol. Horizontal bonds point toward you; vertical bonds point away.**

{"diagramSpecification": {"type": "fischer", "top": "CH□CH□", "bottom": "CH□", "left": "OH", "right": "H", "caption": "Fischer projection of butan-2-ol. Horizontal bonds point toward you; vertical bonds point away."}}

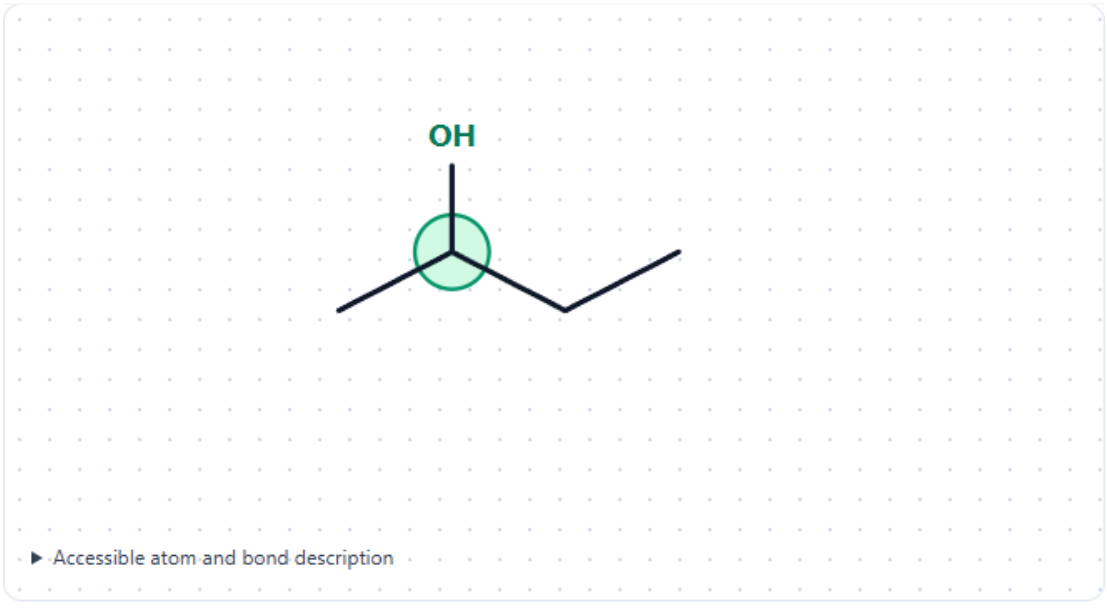
## configuration:step-3 | 3. Mirror the projection

Interchanging H and OH inverts this single center. The mirrored projection is R and represents the enantiomer.

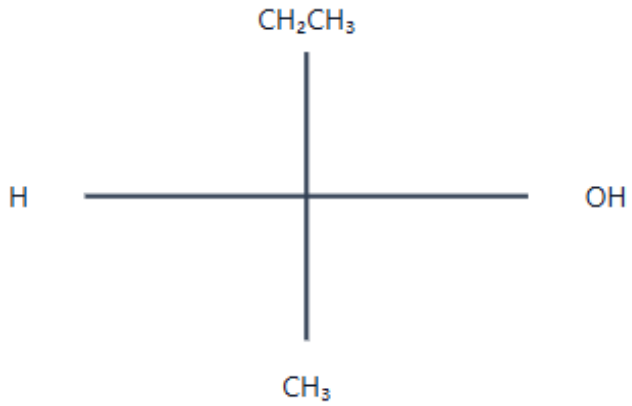
Displayed molecule ID: butanol2

{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: configuration-step-3



Screenshot: configuration-step-3-diagram




**Mirror Fischer projection: exchange the two horizontal groups.**

{"diagramSpecification": {"type": "fischer", "top": "CH□CH□", "bottom": "CH□", "left": "H", "right": "OH", "caption": "Mirror Fischer projection: exchange the two horizontal groups."}}

# alkene | E/Z configuration and optical mixtures

Rank each end separately

## alkene:rule | Core Rules

E/Z is defined only when each double-bond carbon has two different substituents. Rank each end by CIP: Z places the higher-priority groups on the same side; E on opposite sides. A racemate has equal amounts of two enantiomers and zero net rotation under ordinary conditions.

## alkene:cue | Memory Cues

Z: together. E: opposite. Enantiomeric excess is the absolute difference in enantiomer percentages.

## alkene:trap | Common Mistakes / MCAT Traps

Cis/trans cannot replace E/Z in every alkene. Zero optical rotation can mean a racemate or an achiral compound; it does not identify which.

## alkene:app | MCAT Applications

If a pure enantiomer rotates +20° under the same conditions, a 75:25 mixture of it and its enantiomer gives +10° (50% ee).

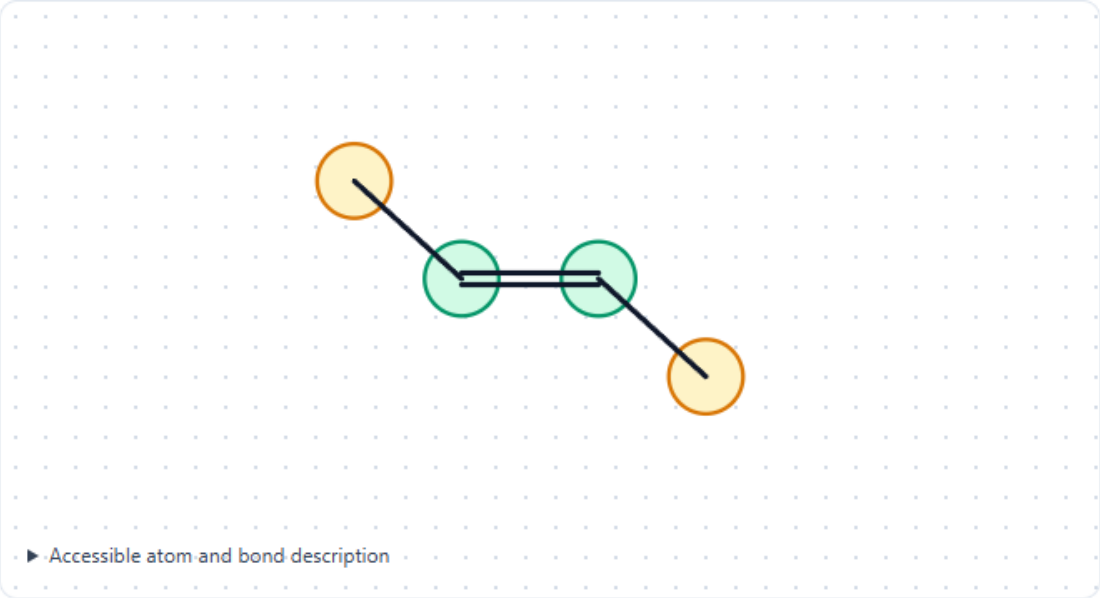
## alkene:step-1 | 1. Compare the two ends

Each alkene carbon bears methyl and H. Methyl outranks H, and the methyl groups lie on opposite sides: E.

Displayed molecule ID: e-butene

{"highlights": {"prin": [1, 2], "sub": [0, 3]}, "revealName": true}

Screenshot: alkene-step-1



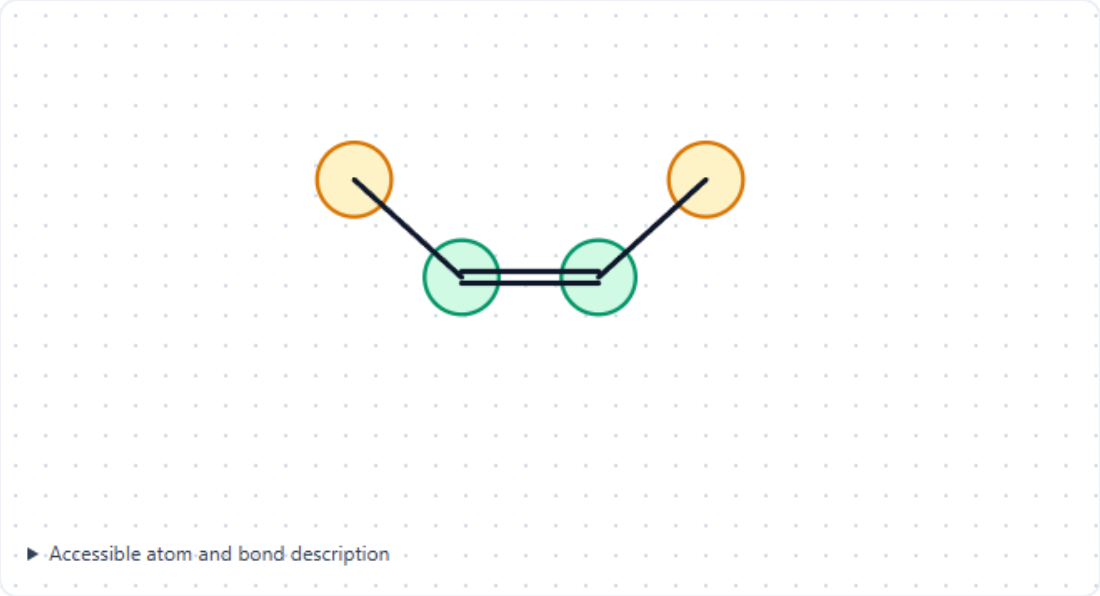

## alkene:step-2 | 2. Same side

Both higher-priority methyl groups now lie on the same side: Z. Rotation about C=C would disrupt pi overlap, so these are configurational isomers.

Displayed molecule ID: z-butene

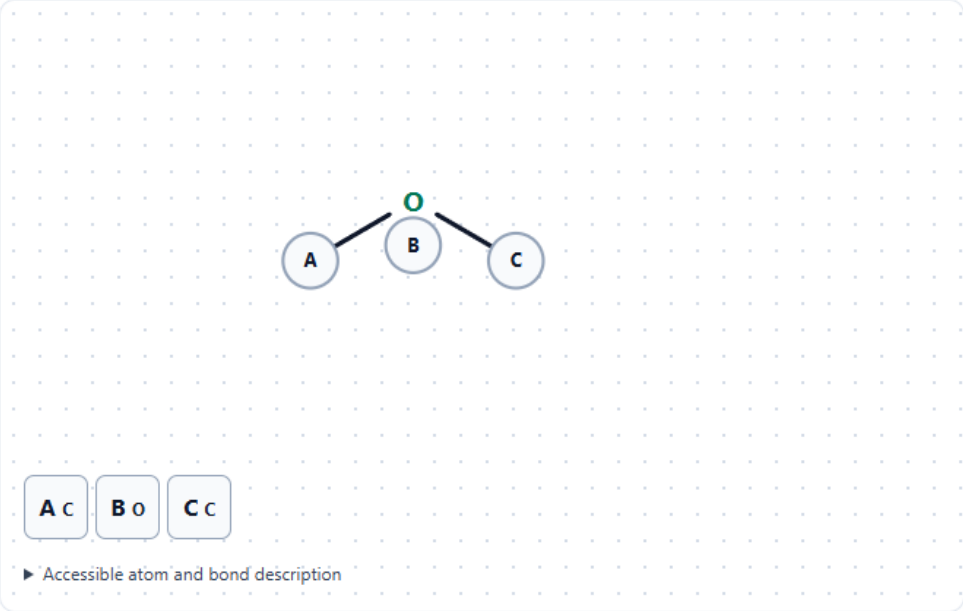
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Screenshot: alkene-step-2



# isomer-pair | Classify an isomer pair

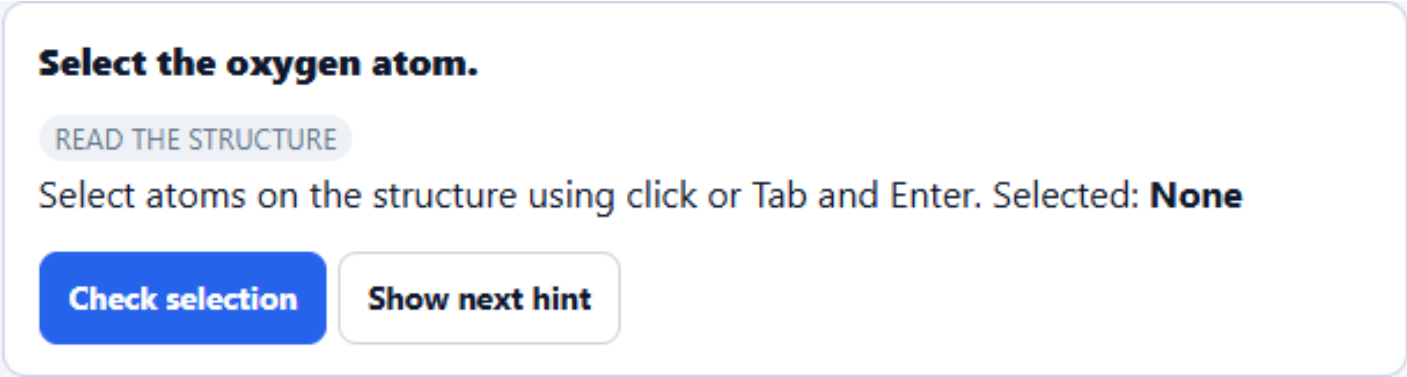
Screenshot: isomer-pair-stage-1-molecule




## isomer-pair:stage-1 | Select the oxygen atom.

Answer type: atoms

Screenshot: isomer-pair-stage-1-question



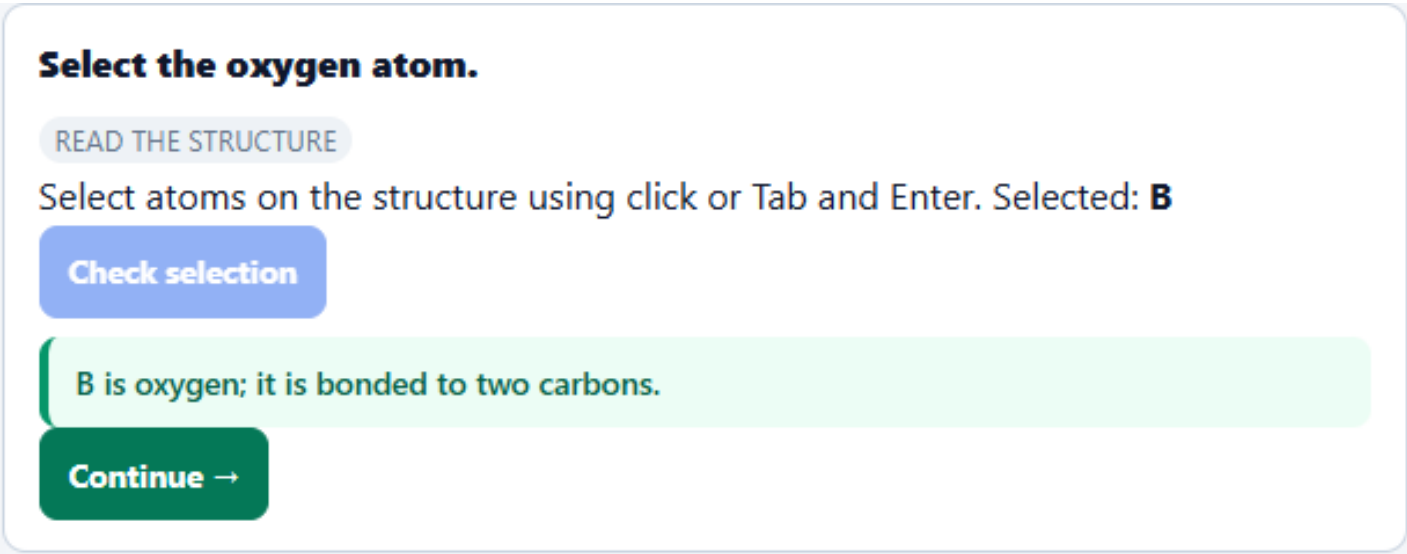

{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1]}}

Hint 1: Read the element label.

Hint 2: The oxygen is between the carbon endpoints.

Explanation: B is oxygen; it is bonded to two carbons.

Screenshot: isomer-pair-stage-1-feedback




### isomer-pair:stage-2 | Compared with ethanol (C2H6O), this compound is a…

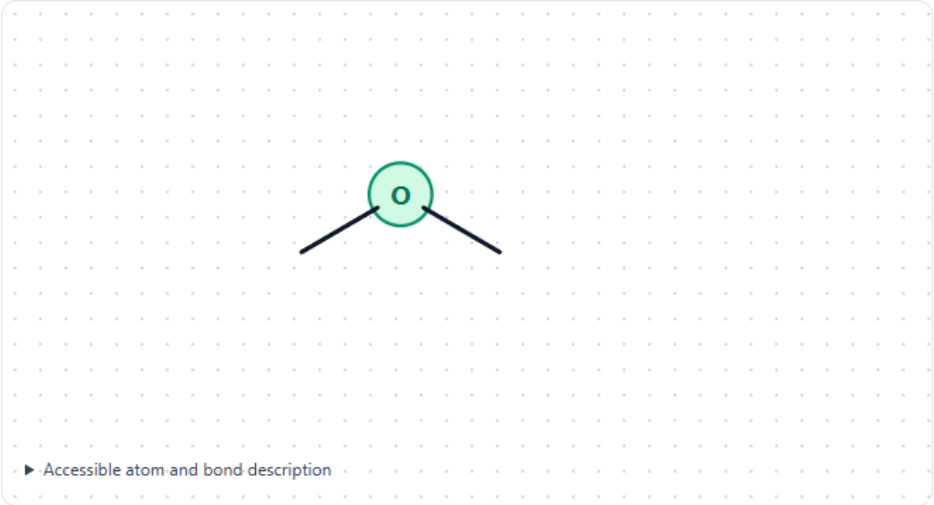
Answer type: choice

Option 0 (A): Constitutional isomer

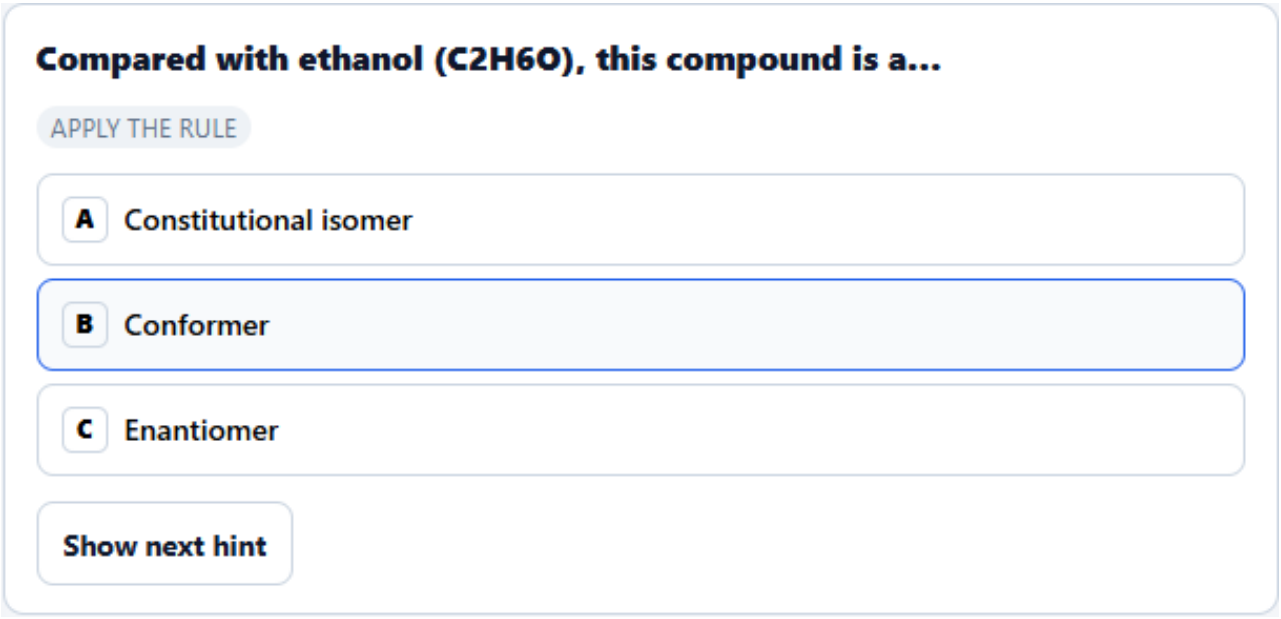
Option 1 (B): Conformer

Option 2 (C): Enantiomer

Screenshot: isomer-pair-stage-2-molecule



Screenshot: isomer-pair-stage-2-question



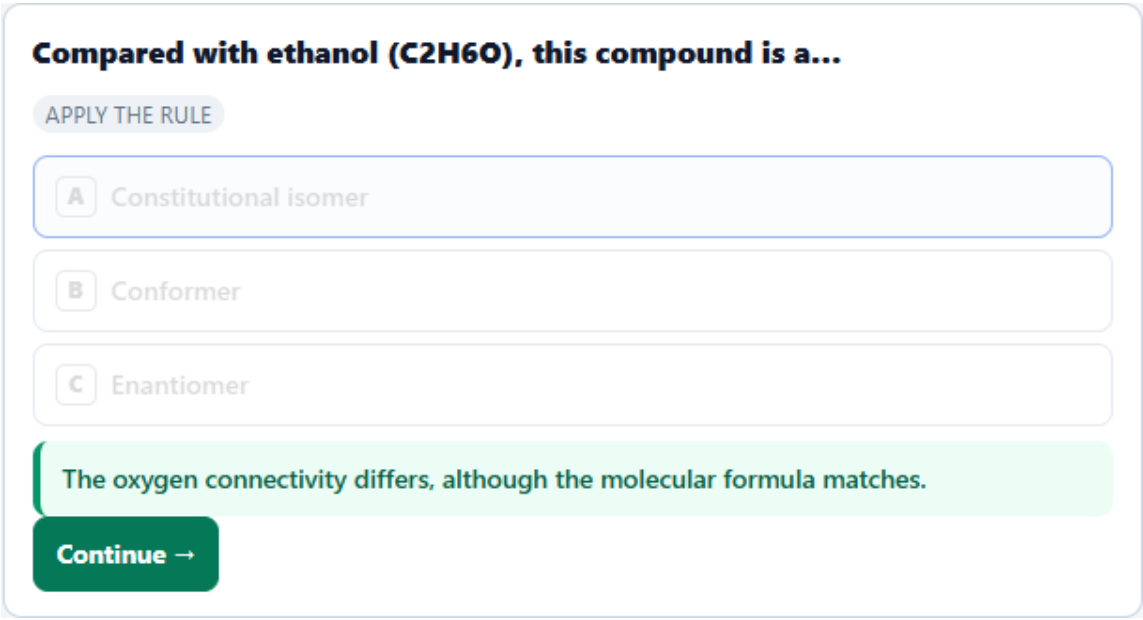

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare bonds before spatial arrangements.

Hint 2: Ethanol contains an O–H bond.

Explanation: The oxygen connectivity differs, although the molecular formula matches.

Screenshot: isomer-pair-stage-2-feedback




## isomer-pair:stage-3 | Which can donate an O–H hydrogen bond?

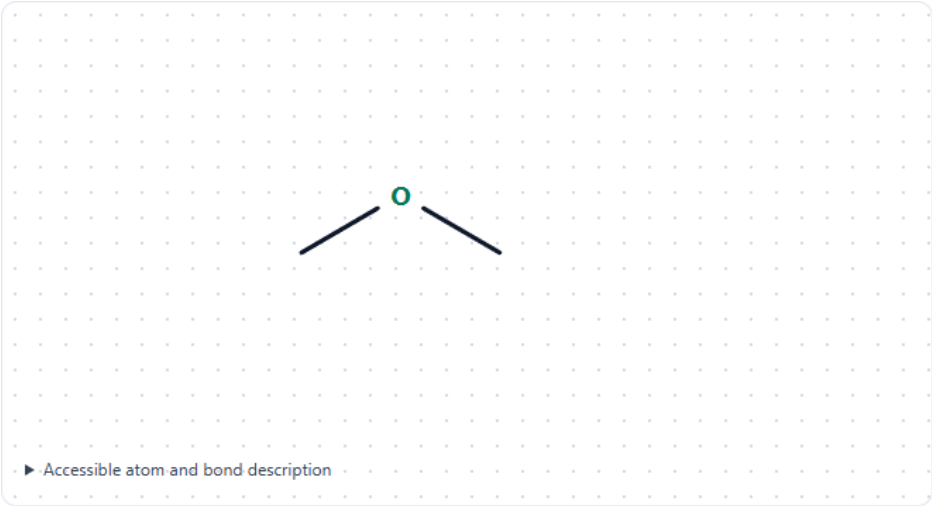
Answer type: choice

Option 0 (A): Both

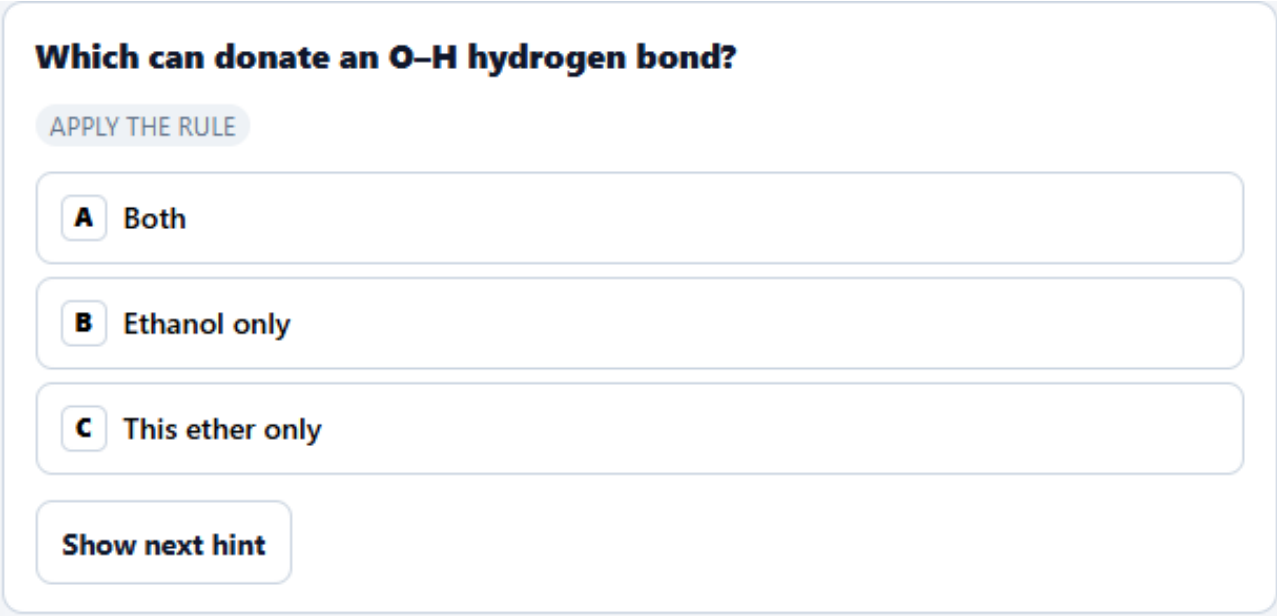
Option 1 (B): Ethanol only

Option 2 (C): This ether only

Screenshot: isomer-pair-stage-3-molecule




Screenshot: isomer-pair-stage-3-question



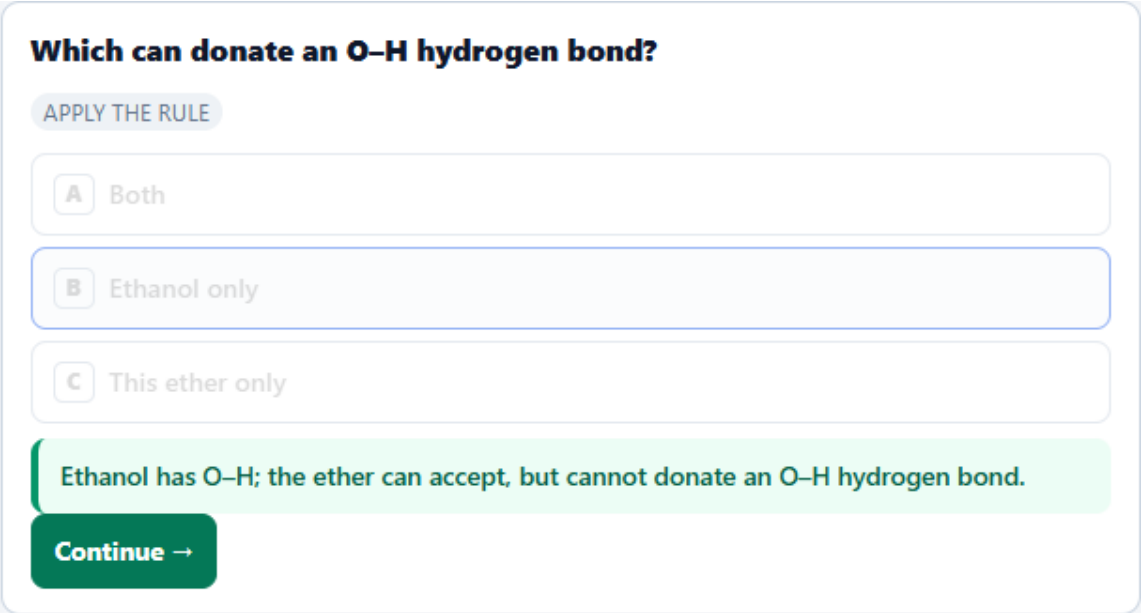

{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Donation requires H bonded to an electronegative atom.

Hint 2: Look for an O–H bond.

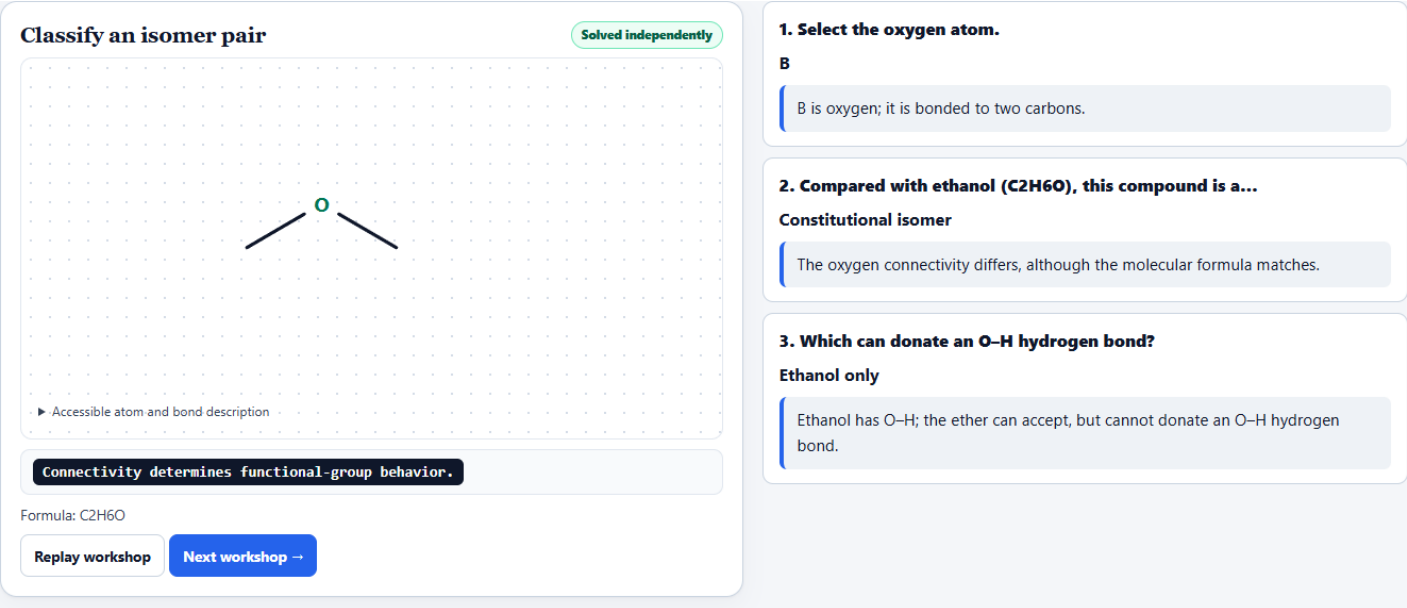
Explanation: Ethanol has O–H; the ether can accept, but cannot donate an O–H hydrogen bond.

Screenshot: isomer-pair-stage-3-feedback



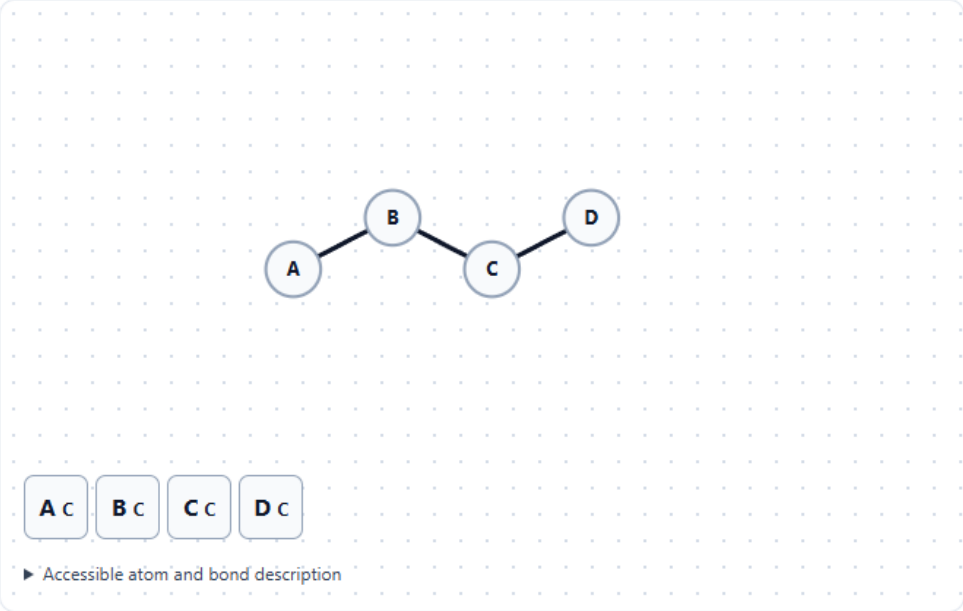

Completion summary: Connectivity determines functional-group behavior.

Screenshot: isomer-pair-complete

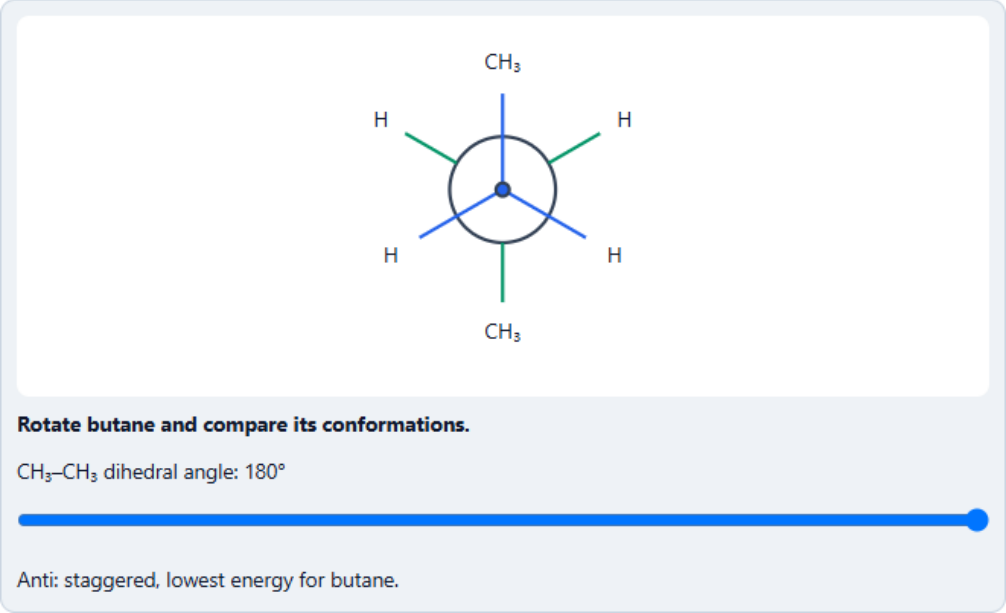


# butane-rotation | Reason about conformational energy

Screenshot: butane-rotation-stage-1-molecule




Screenshot: butane-rotation-diagram

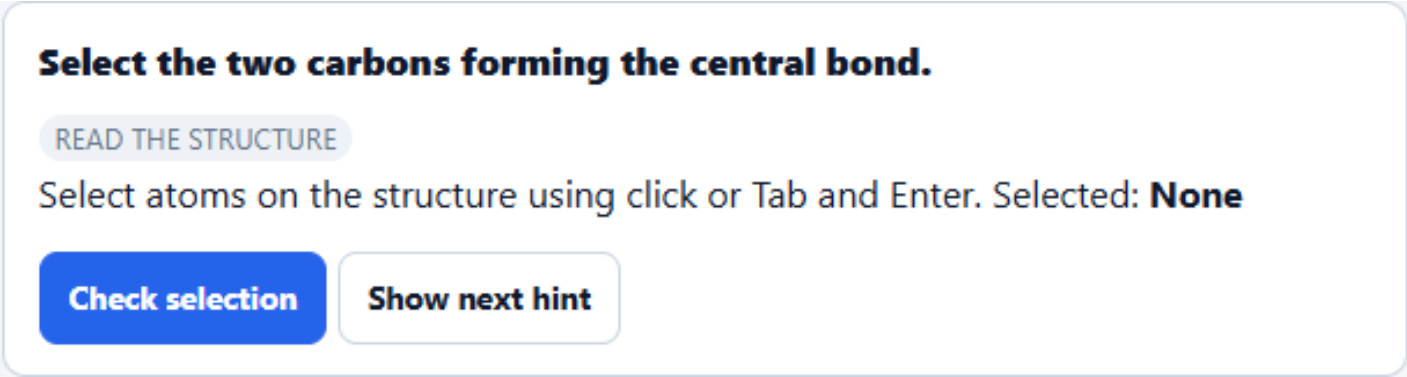



{"diagramSpecification": {"type": "newman", "caption": "Rotate butane and compare its conformations."}}

## butane-rotation:stage-1 | Select the two carbons forming the central bond.

Answer type: atoms

Screenshot: butane-rotation-stage-1-question



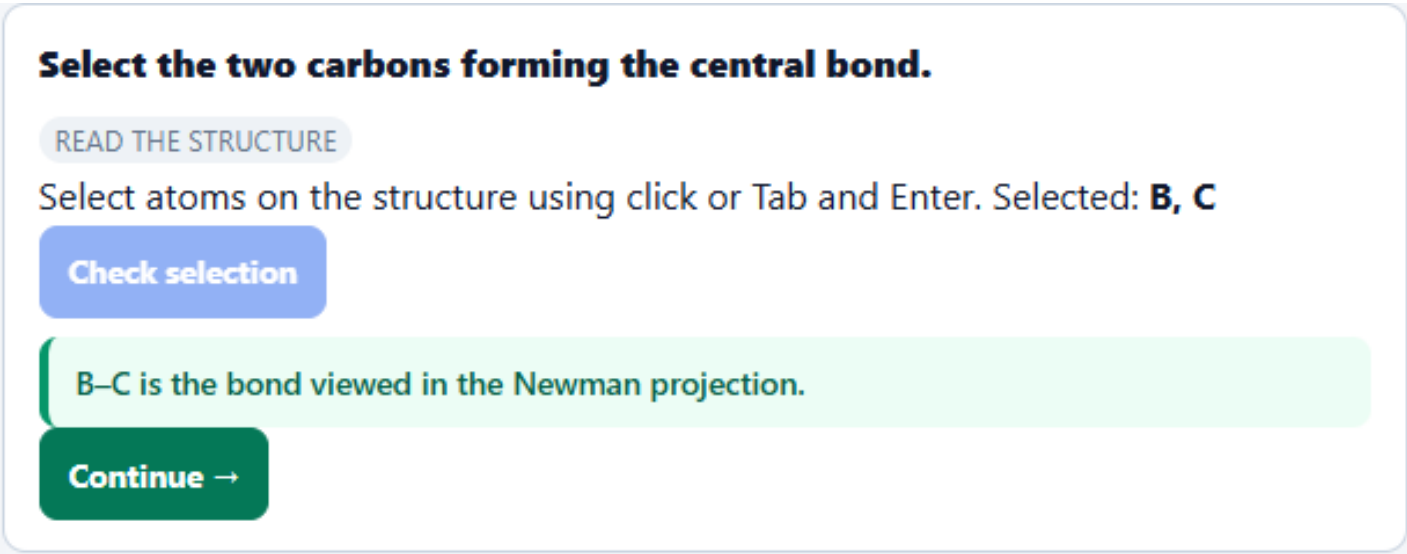

{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Find the four-carbon chain.

Hint 2: Choose the two internal atoms.

Explanation: B–C is the bond viewed in the Newman projection.

Screenshot: butane-rotation-stage-1-feedback




### butane-rotation:stage-2 | At a 180° methyl–methyl dihedral, the conformation is…

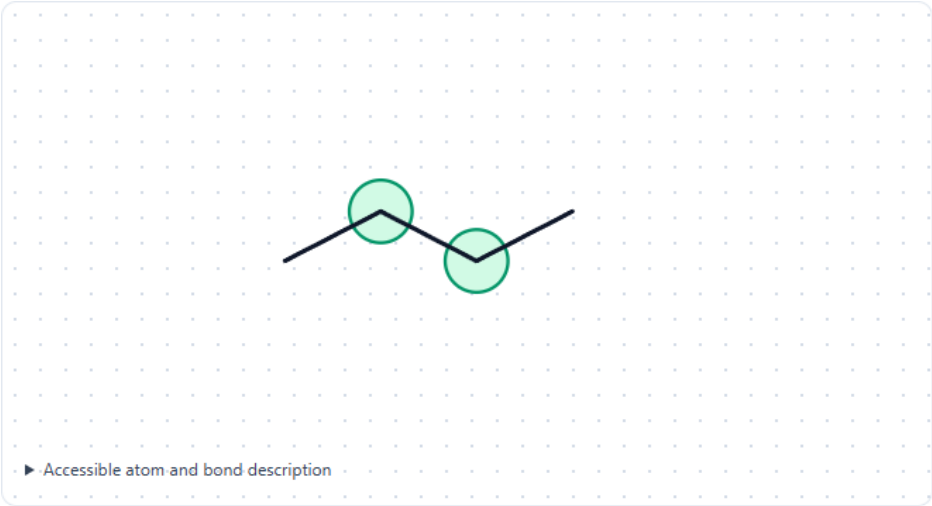
Answer type: choice

Option 0 (A): Anti

Option 1 (B): Gauche

Option 2 (C): Fully eclipsed

Screenshot: butane-rotation-stage-2-molecule




Screenshot: butane-rotation-stage-2-question



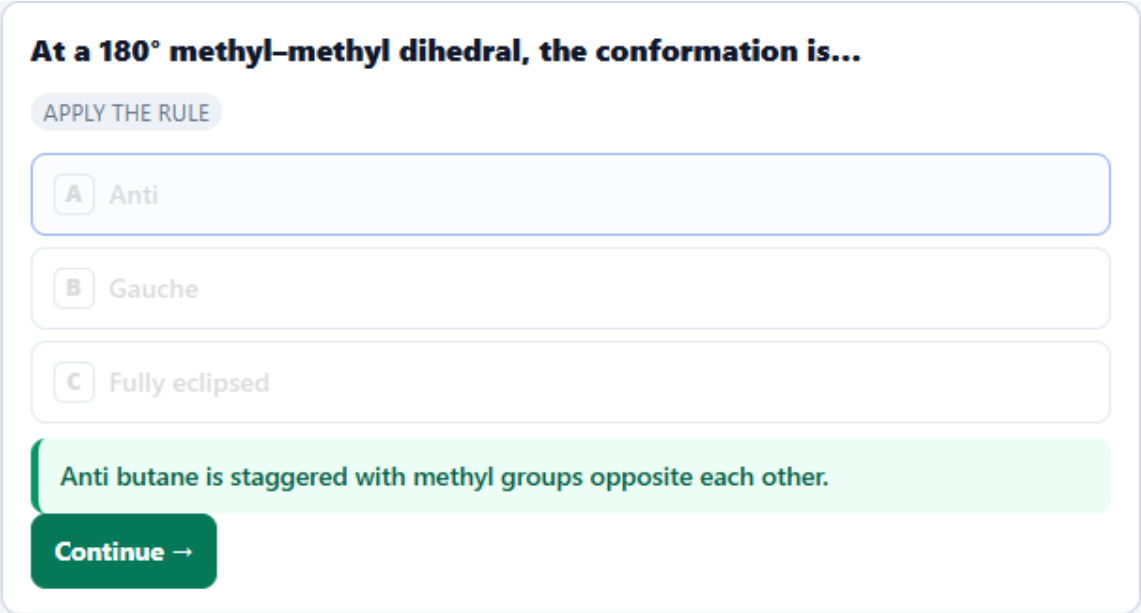

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish staggered from eclipsed.

Hint 2: Opposite groups are separated by half a turn.

Explanation: Anti butane is staggered with methyl groups opposite each other.

Screenshot: butane-rotation-stage-2-feedback




## butane-rotation:stage-3 | A cyclohexane chair flip changes…

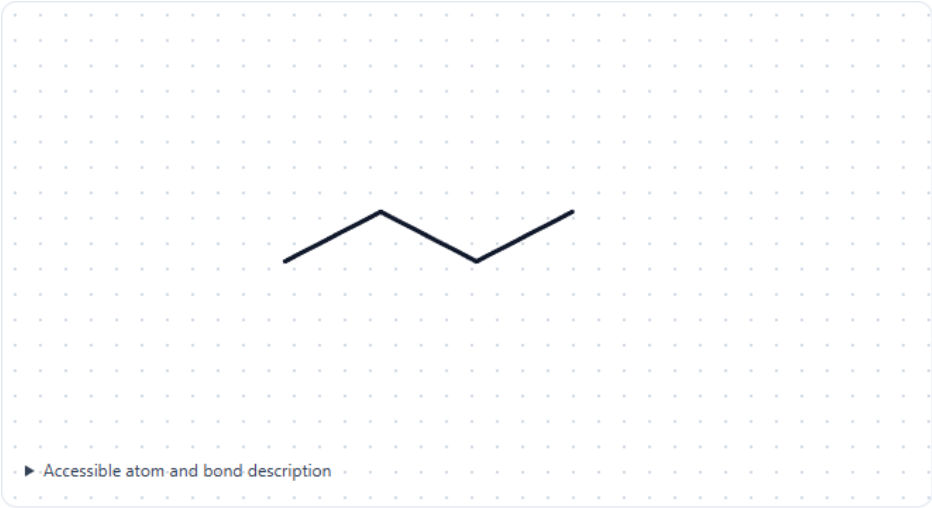
Answer type: choice

Option 0 (A): Up into down

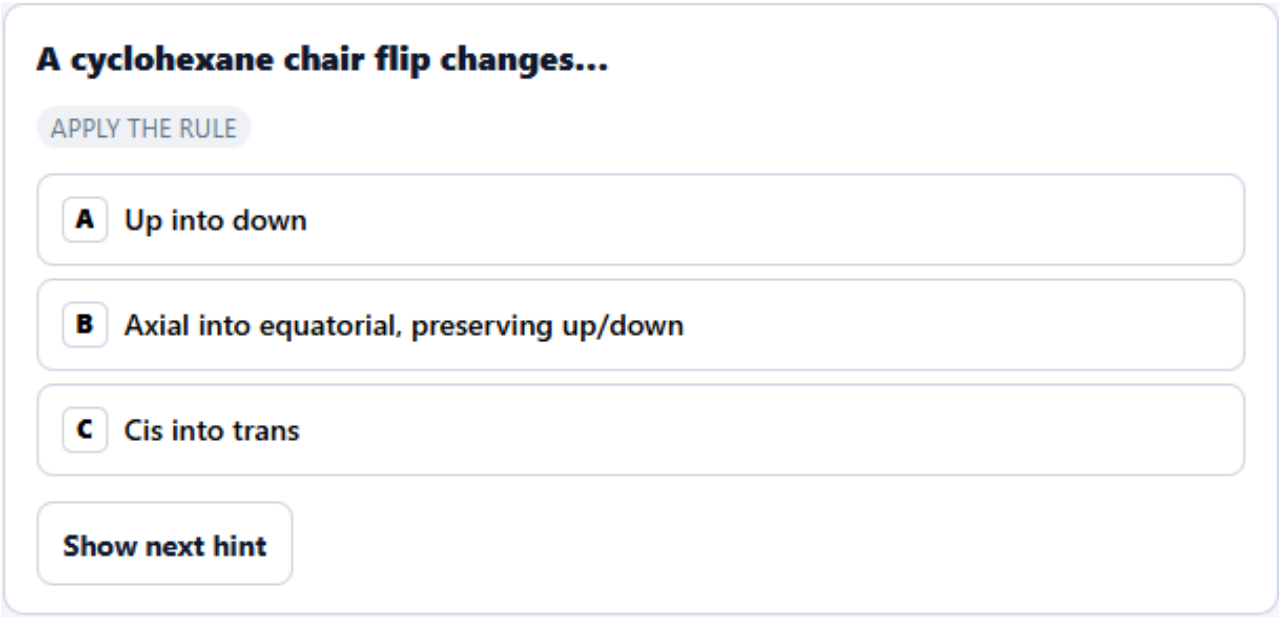
Option 1 (B): Axial into equatorial, preserving up/down

Option 2 (C): Cis into trans

Screenshot: butane-rotation-stage-3-molecule



Screenshot: butane-rotation-stage-3-question



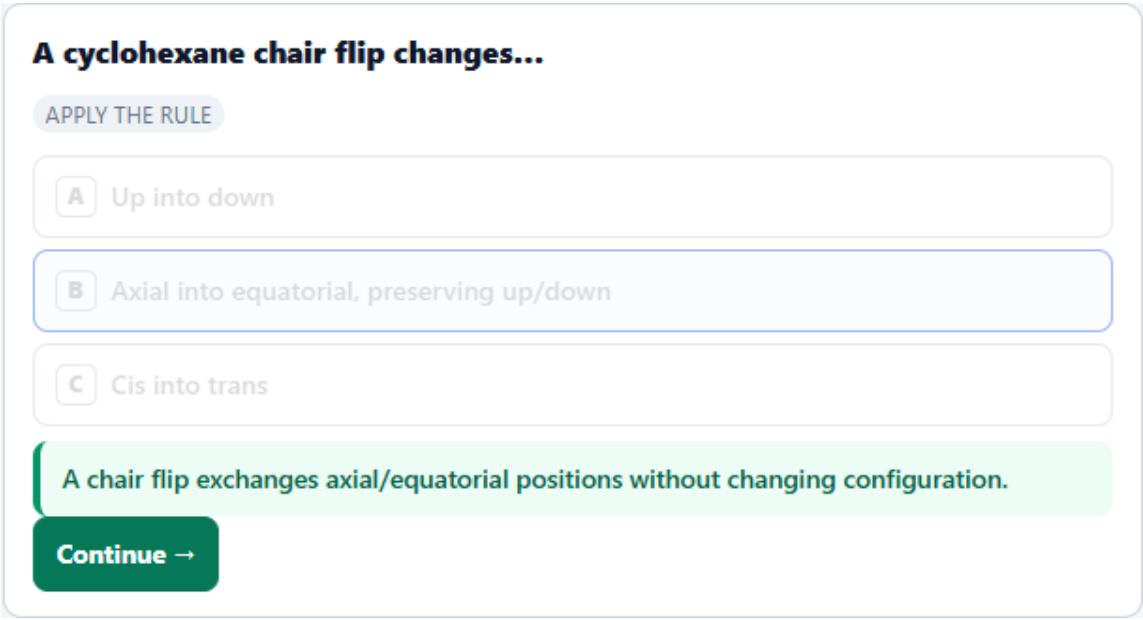

{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Track the bond orientation relative to the ring.

Hint 2: Up/down and axial/equatorial are different labels.

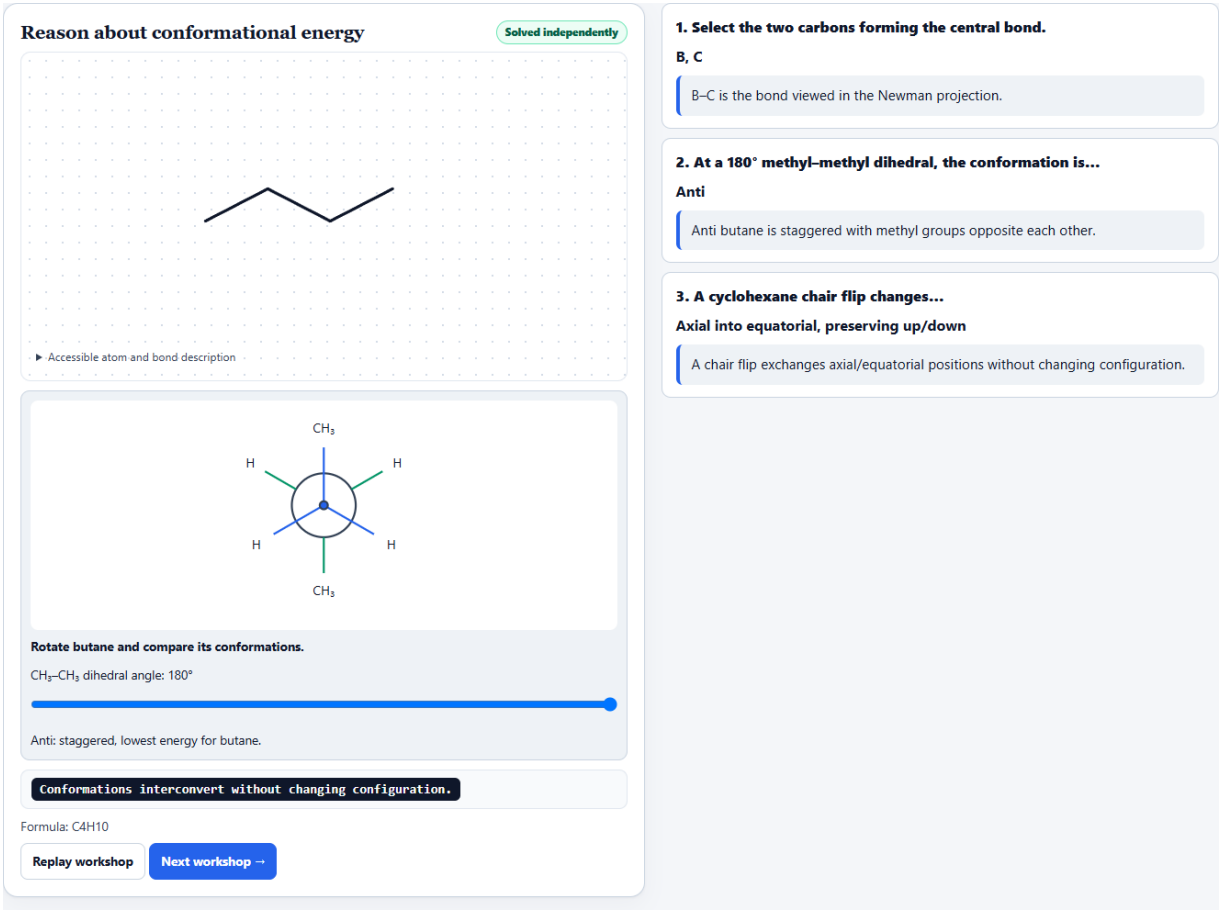
Explanation: A chair flip exchanges axial/equatorial positions without changing configuration.

Screenshot: butane-rotation-stage-3-feedback



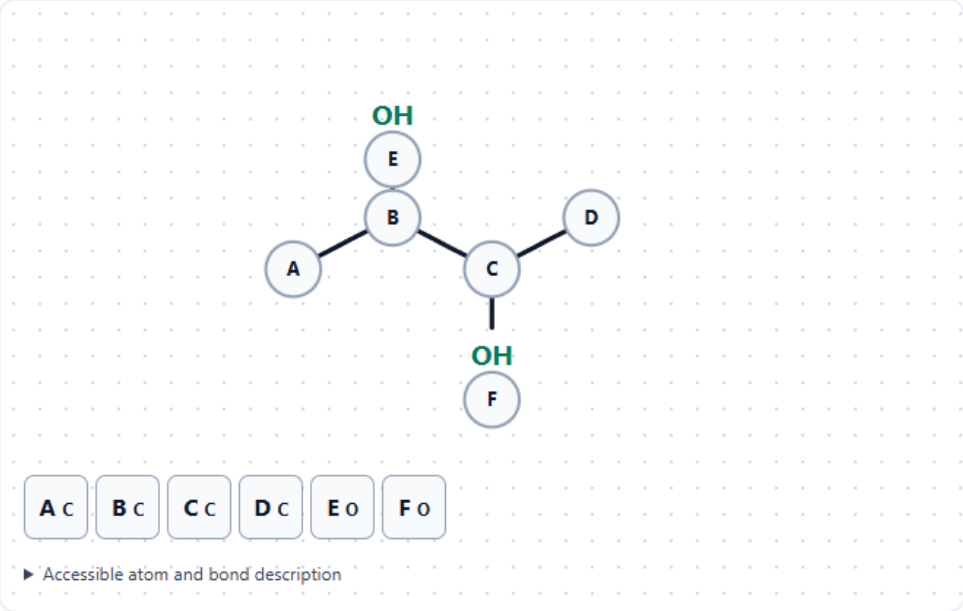

Completion summary: Conformations interconvert without changing configuration.

Screenshot: butane-rotation-complete



# stereocenters | Count stereoisomers with symmetry

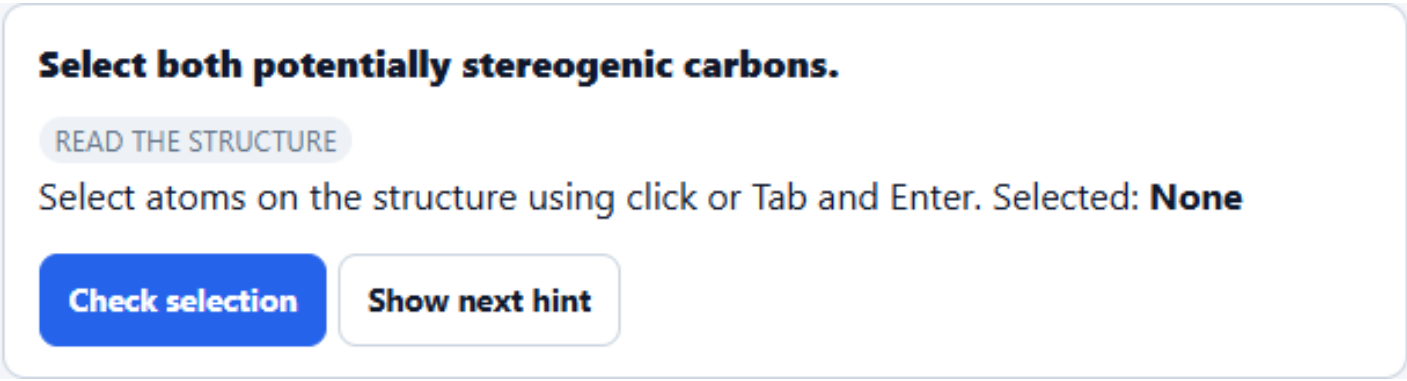
Screenshot: stereocenters-stage-1-molecule




## stereocenters:stage-1 | Select both potentially stereogenic carbons.

Answer type: atoms

Screenshot: stereocenters-stage-1-question



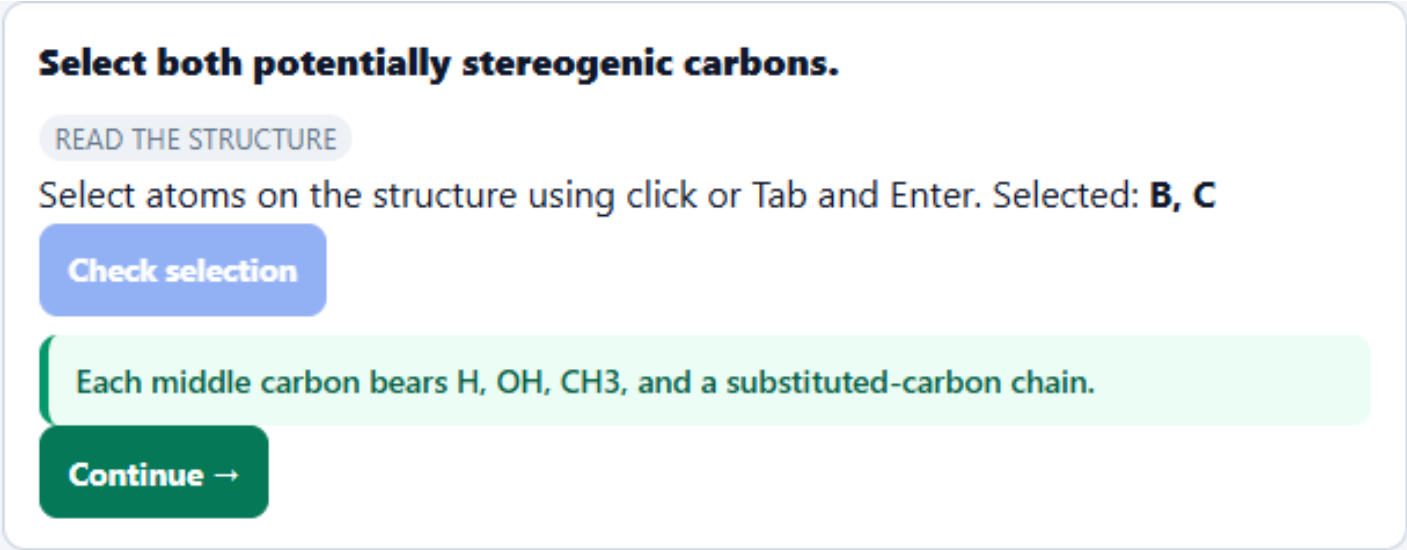

{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Look for tetrahedral atoms with four different groups.

Hint 2: The OH-bearing carbons are the candidates.

Explanation: Each middle carbon bears H, OH, CH3, and a substituted-carbon chain.

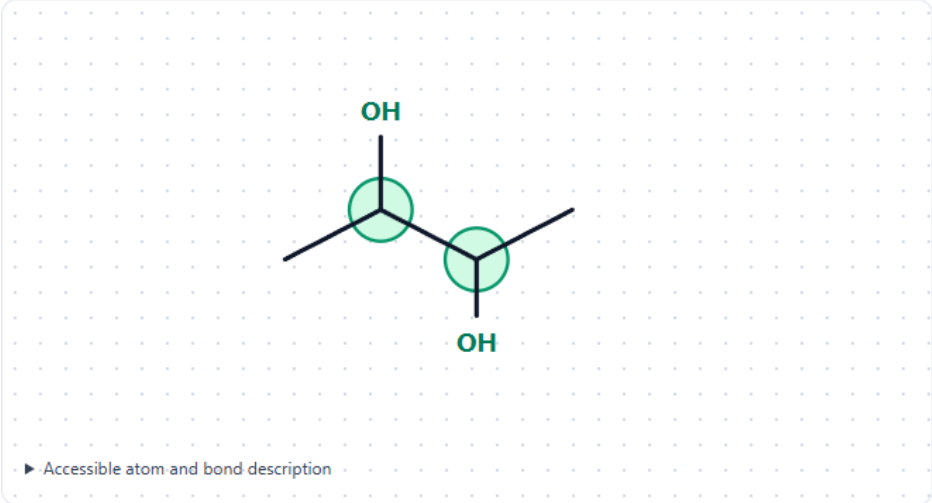
Screenshot: stereocenters-stage-1-feedback



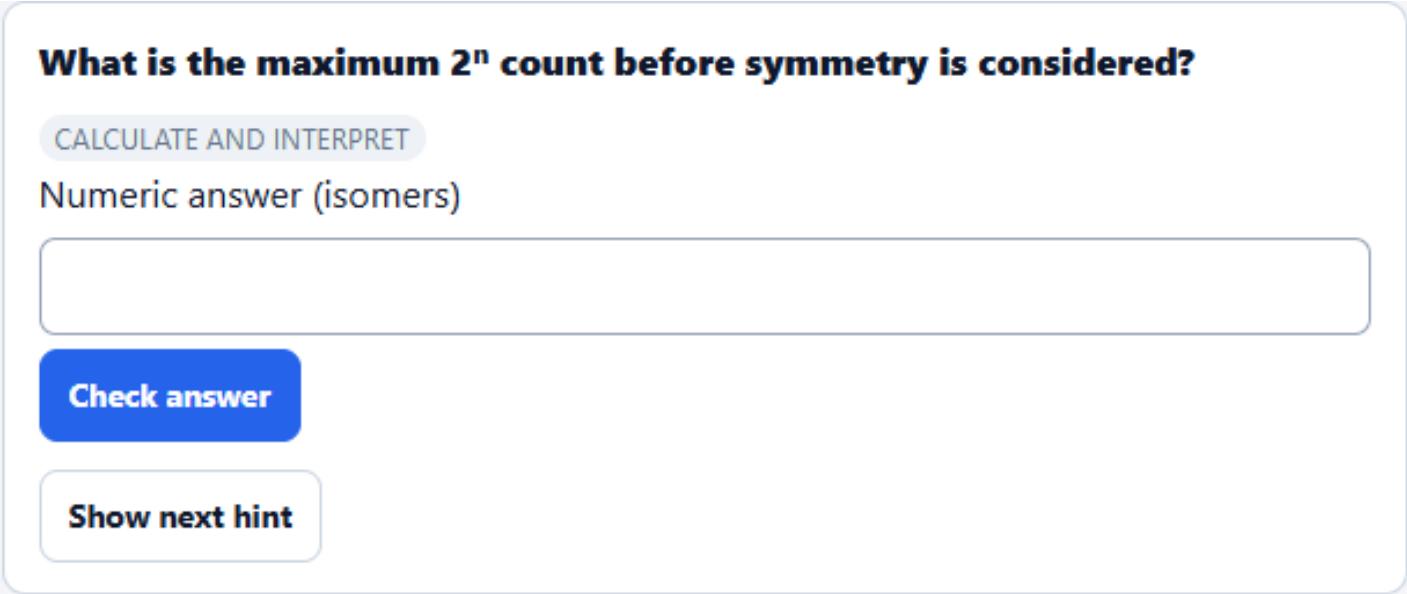

### stereocenters:stage-2 | What is the maximum $2^n$ count before symmetry is considered?

Answer type: number | unit: isomers

Screenshot: stereocenters-stage-2-molecule



Screenshot: stereocenters-stage-2-question



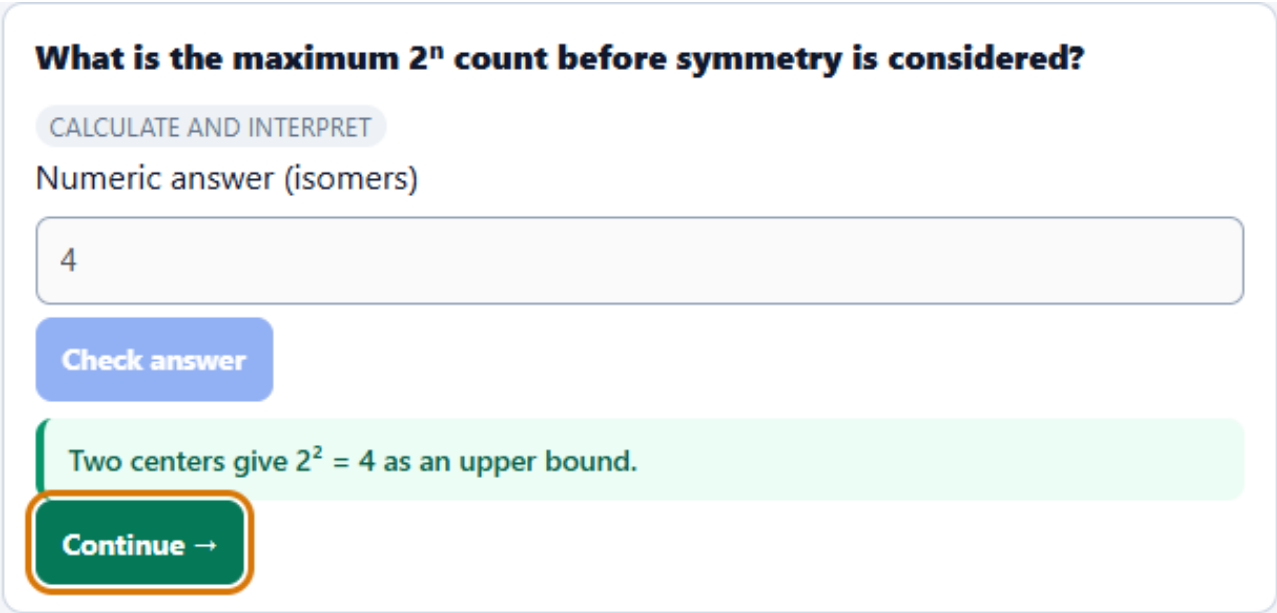

{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: n is the number of independent stereocenters.

Hint 2: Raise two to the power of two.

Explanation: Two centers give $2^2 = 4$ as an upper bound.

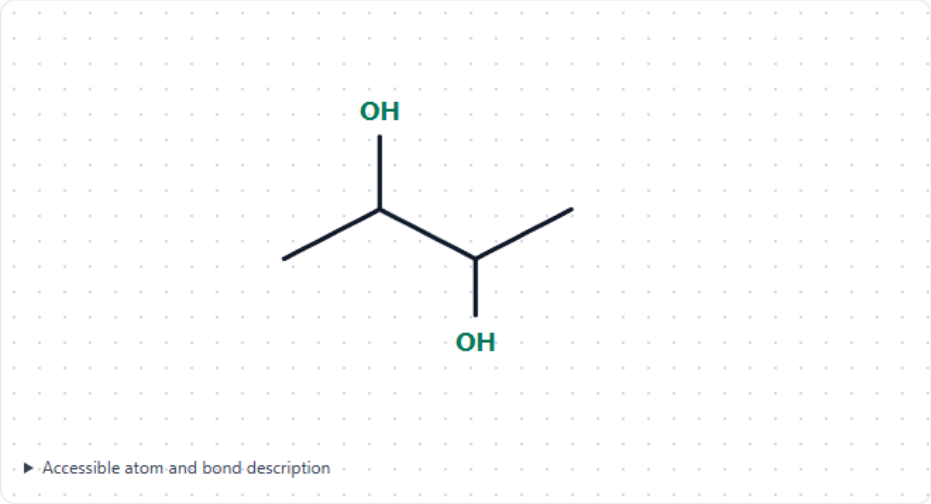
Screenshot: stereocenters-stage-2-feedback



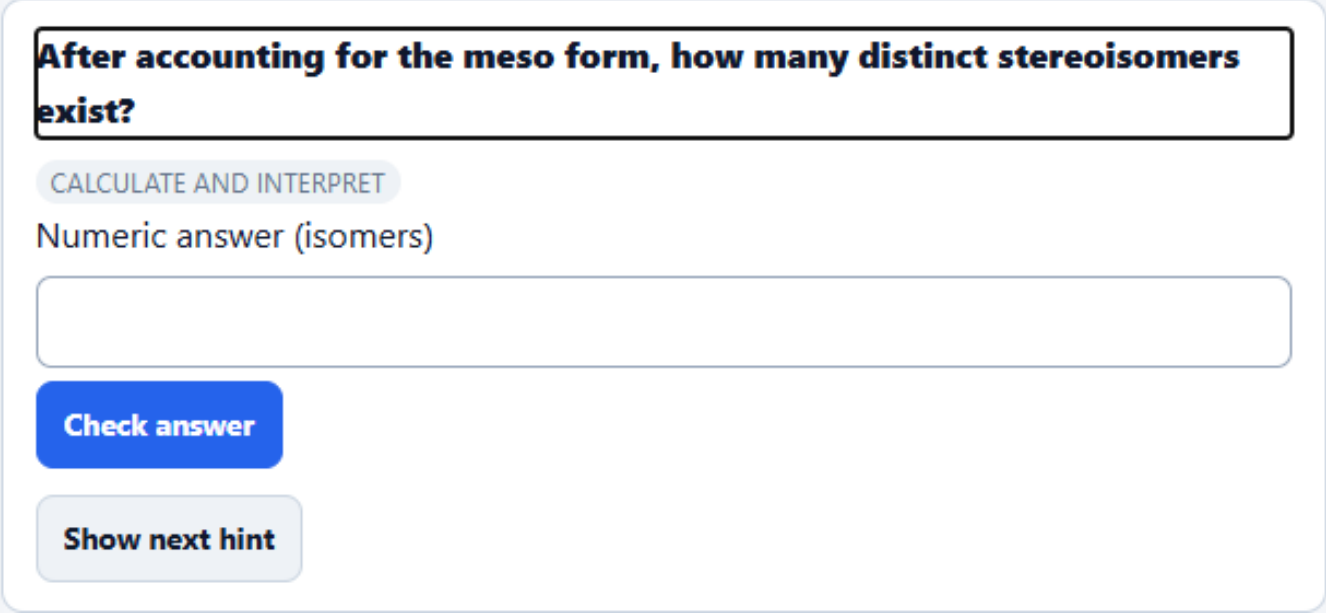

### stereocenters:stage-3 | After accounting for the meso form, how many distinct stereoisomers exist?

Answer type: number | unit: isomers

Screenshot: stereocenters-stage-3-molecule




Screenshot: stereocenters-stage-3-question



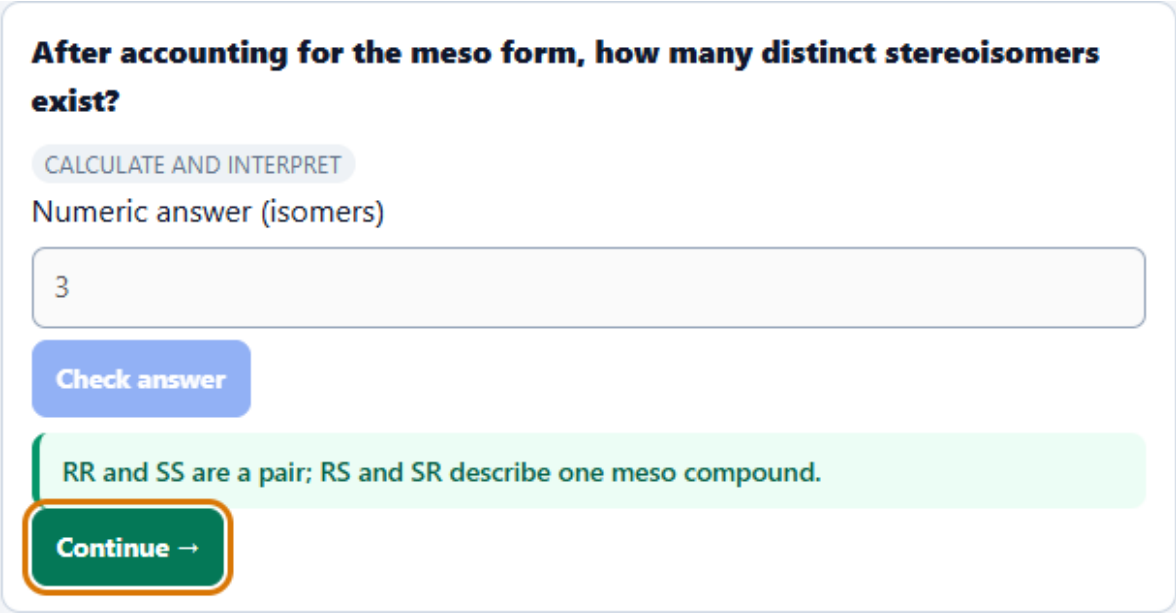

{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Check whether exchanging the equivalent ends duplicates a structure.

Hint 2: There is one achiral meso form plus one mirror-image pair.

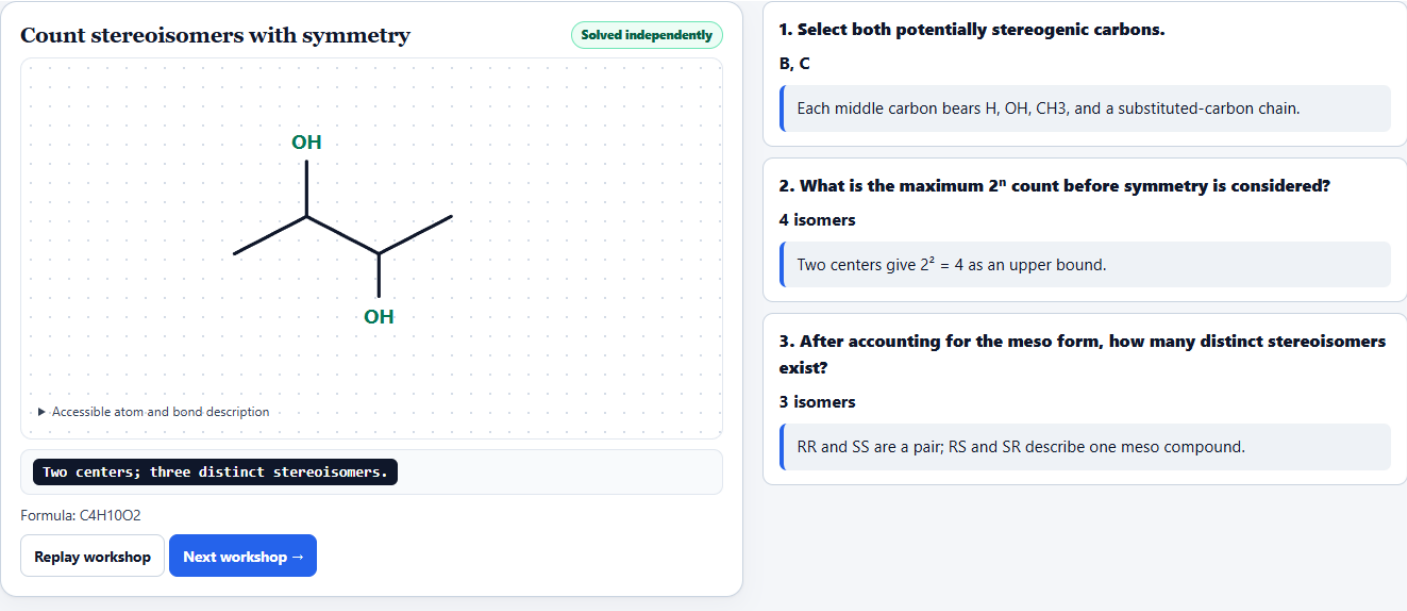
Explanation: RR and SS are a pair; RS and SR describe one meso compound.

Screenshot: stereocenters-stage-3-feedback



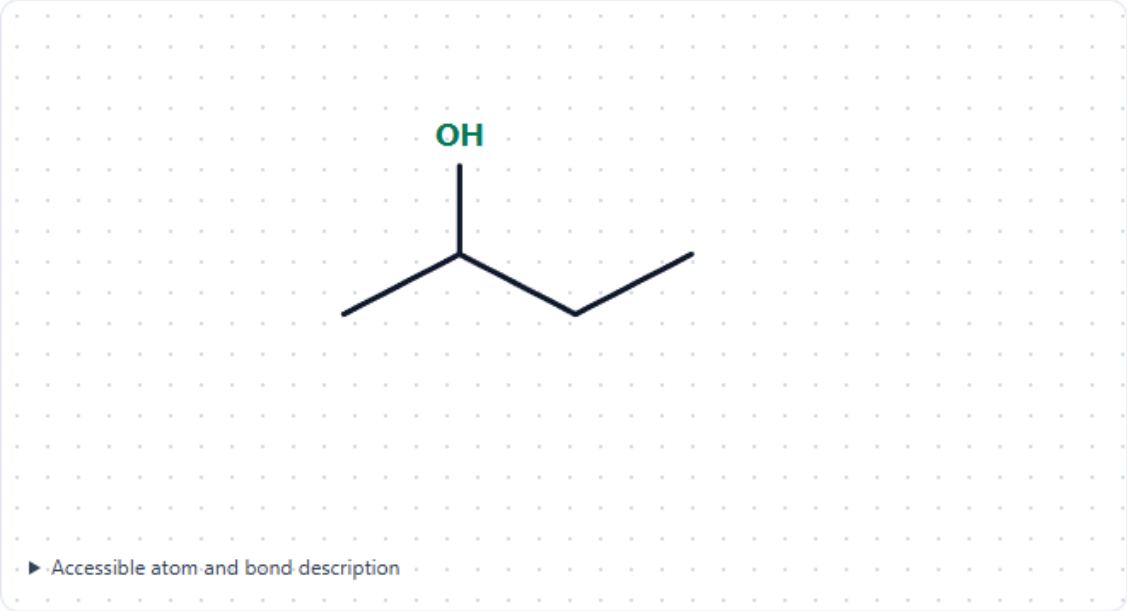

Completion summary: Two centers; three distinct stereoisomers.

Screenshot: stereocenters-complete

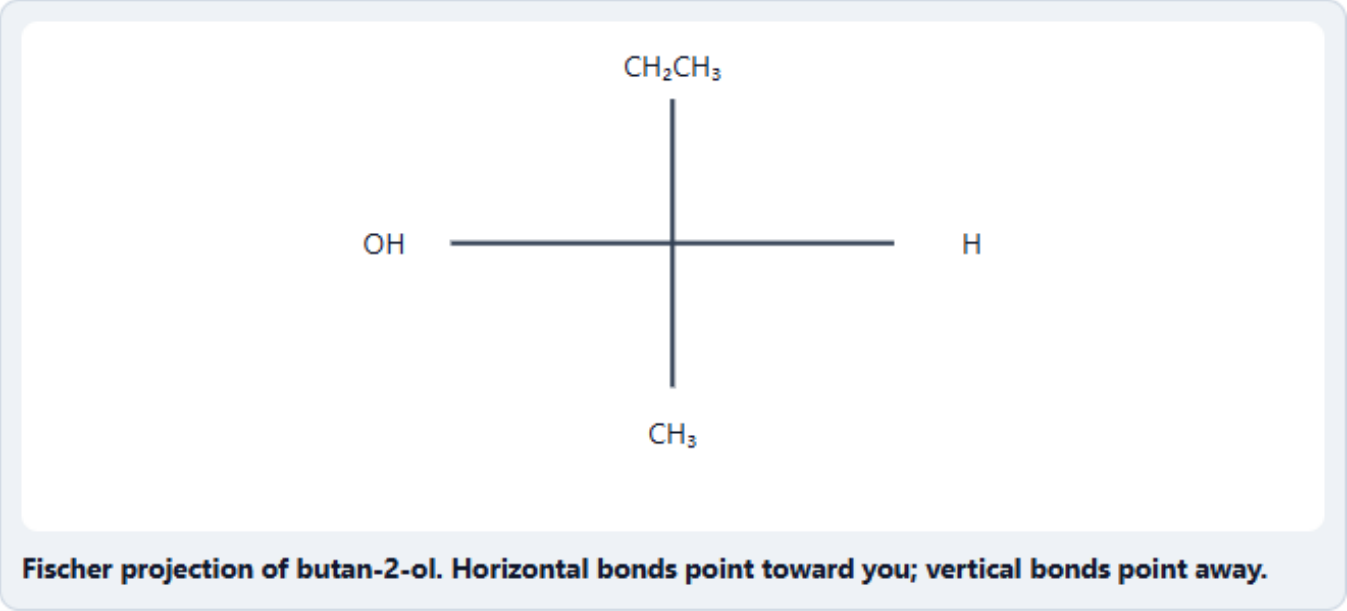


# fischer-cip | Assign a Fischer configuration

Screenshot: fischer-cip-stage-1-molecule




Screenshot: fischer-cip-diagram




{"diagramSpecification": {"type": "fischer", "top": "CH□CH□", "bottom": "CH□", "left": "OH", "right": "H", "caption": "Fischer projection of butan-2-ol. Horizontal bonds point toward you; vertical bonds point away."}}

## fischer-cip:stage-1 | Which substituent has highest CIP priority?

Answer type: choice

Option 0 (A): OH

Option 1 (B): Ethyl

Option 2 (C): Methyl

Option 3 (D): H

Screenshot: fischer-cip-stage-1-question



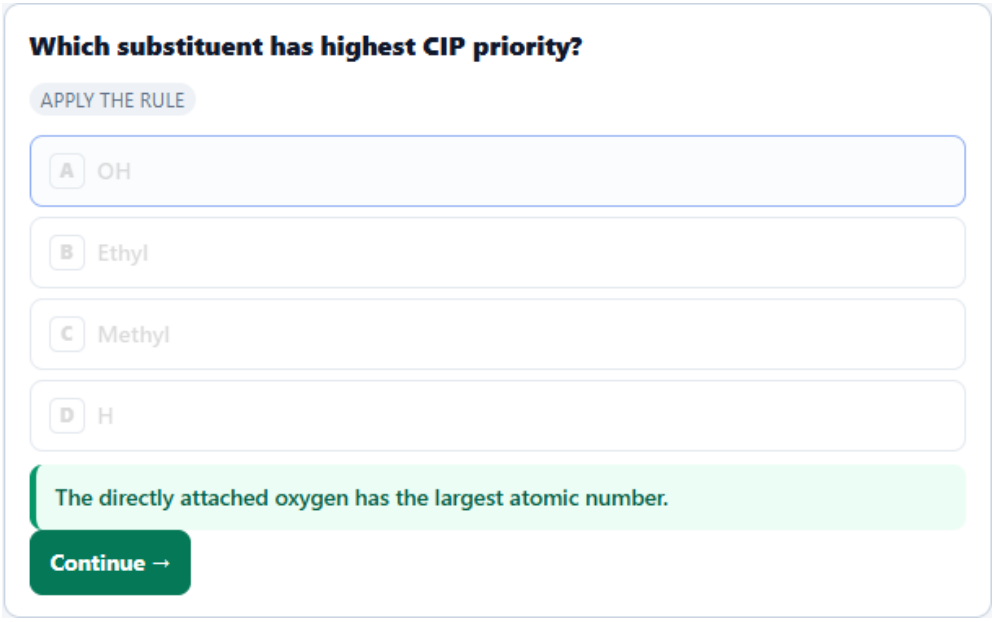

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Start at directly attached atoms.

Hint 2: O has atomic number 8; C is 6.

Explanation: The directly attached oxygen has the largest atomic number.

Screenshot: fischer-cip-stage-1-feedback




## fischer-cip:stage-2 | In this projection, the lowest-priority H points…

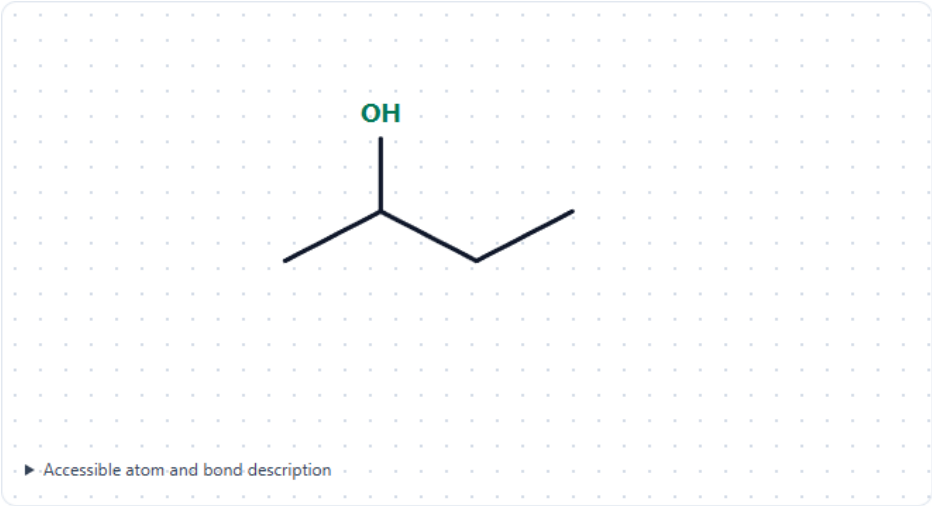
Answer type: choice

Option 0 (A): Away

Option 1 (B): Toward you

Option 2 (C): In the paper plane

Screenshot: fischer-cip-stage-2-molecule




Screenshot: fischer-cip-stage-2-question



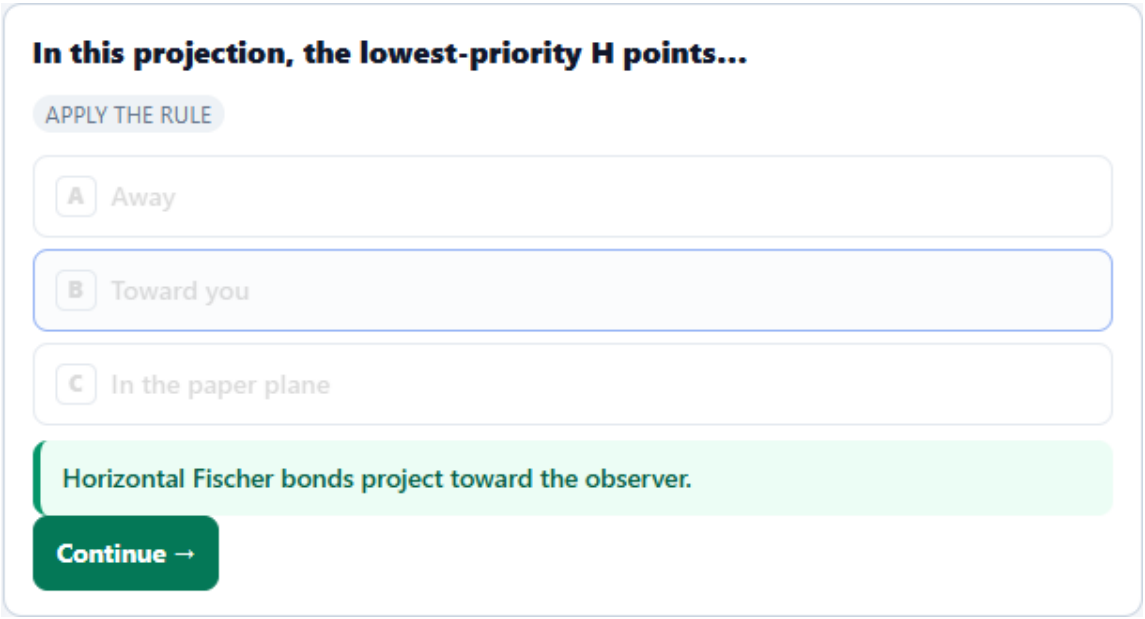

{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the Fischer convention.

Hint 2: Vertical and horizontal bonds have opposite depth directions.

Explanation: Horizontal Fischer bonds project toward the observer.

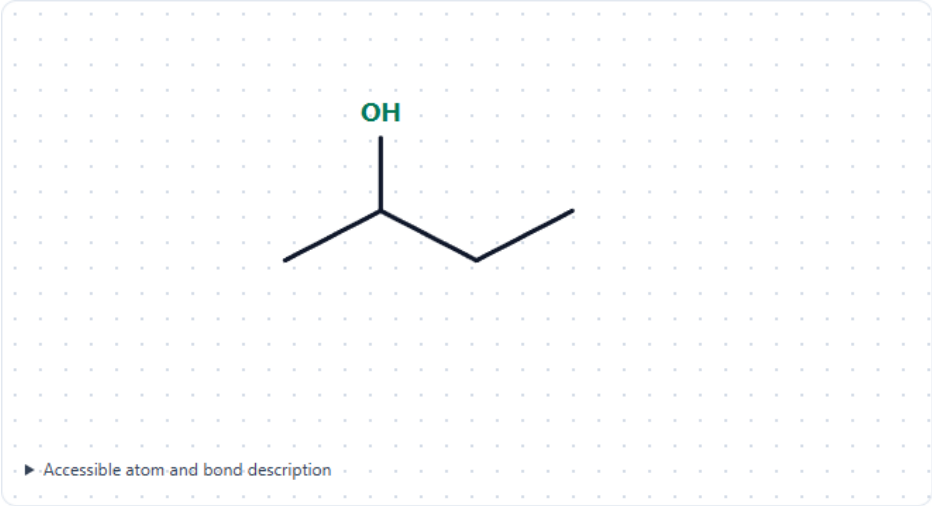
Screenshot: fischer-cip-stage-2-feedback



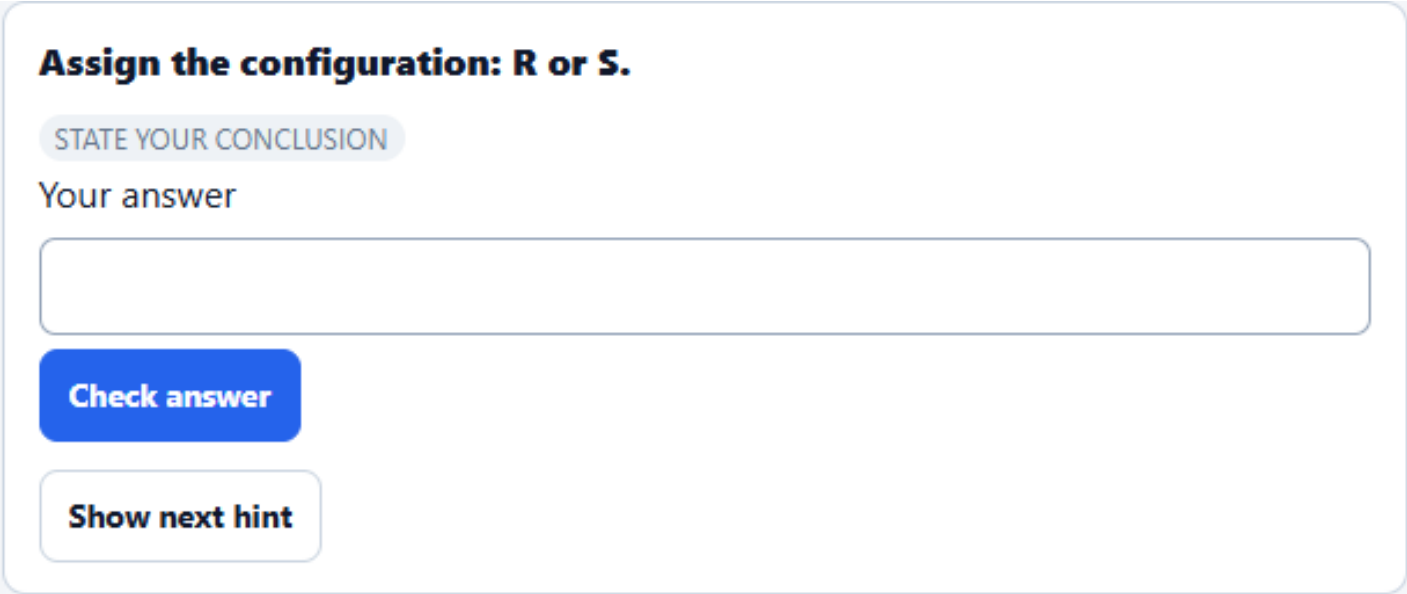

## fischer-cip:stage-3 | Assign the configuration: R or S.

Answer type: text

Screenshot: fischer-cip-stage-3-molecule



Screenshot: fischer-cip-stage-3-question



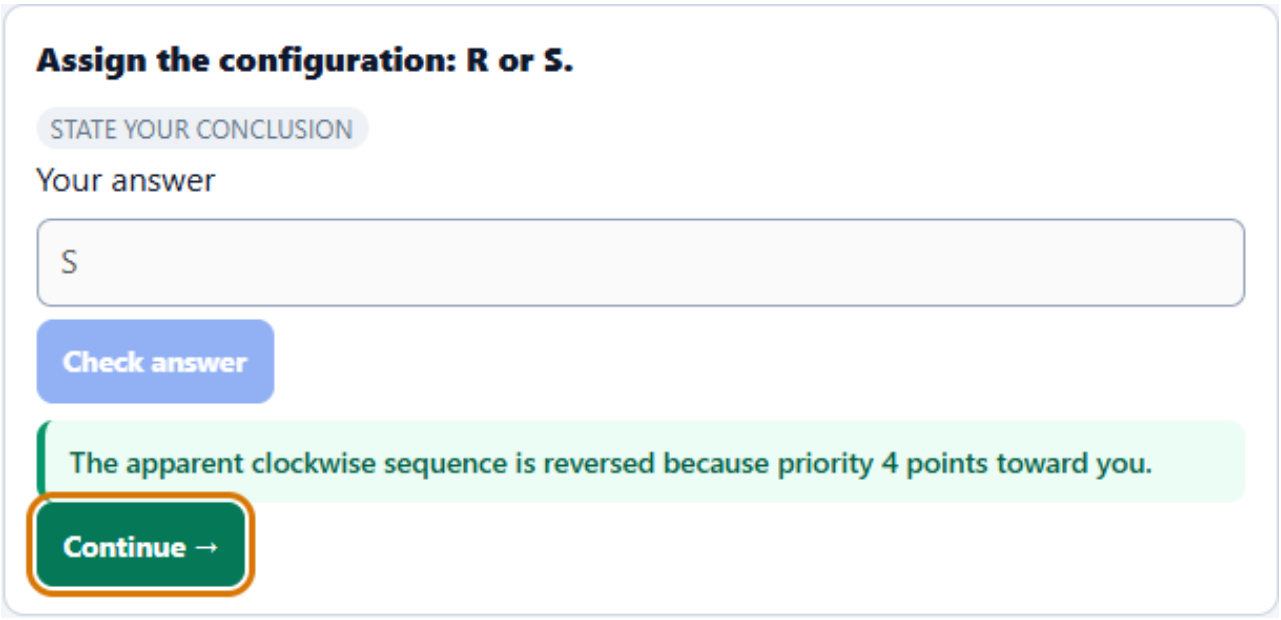

{"authorAnswer": ["S", "(S)"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rank OH > ethyl > methyl > H.

Hint 2: Trace 1→2→3, then reverse because H is horizontal.

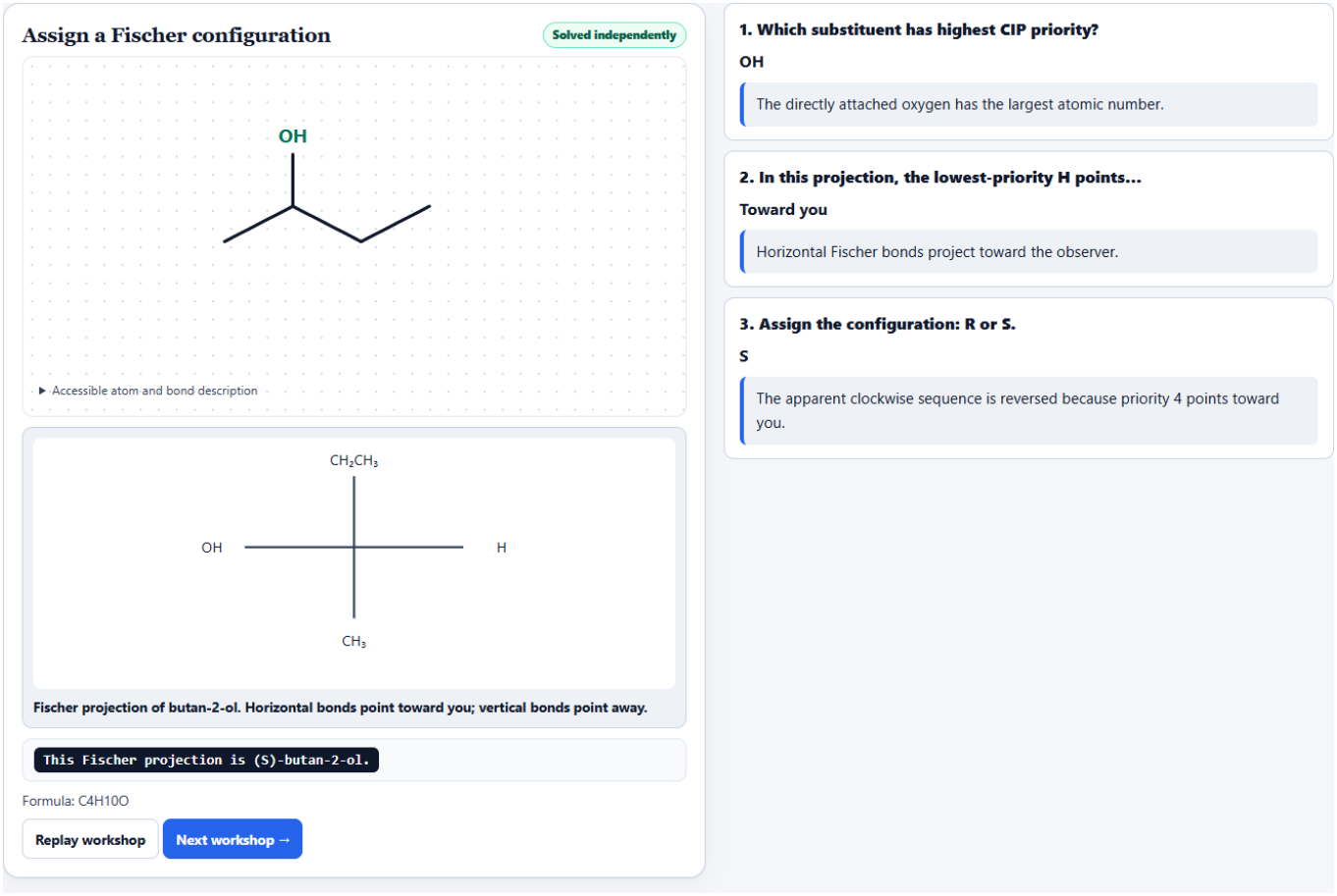
Explanation: The apparent clockwise sequence is reversed because priority 4 points toward you.

Screenshot: fischer-cip-stage-3-feedback



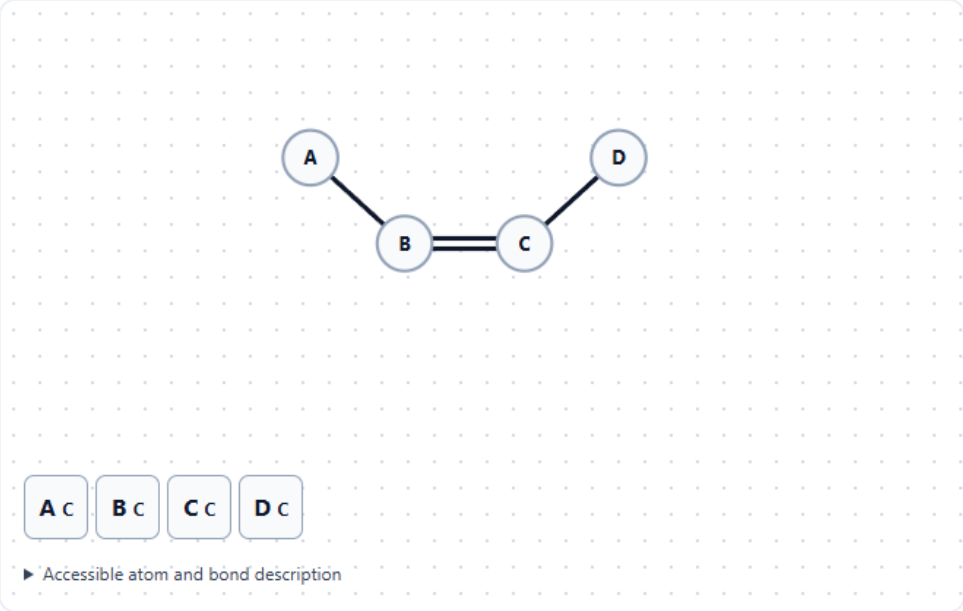

Completion summary: This Fischer projection is (S)-butan-2-ol.

Screenshot: fischer-cip-complete



# ez-ee | Separate geometry from optical activity

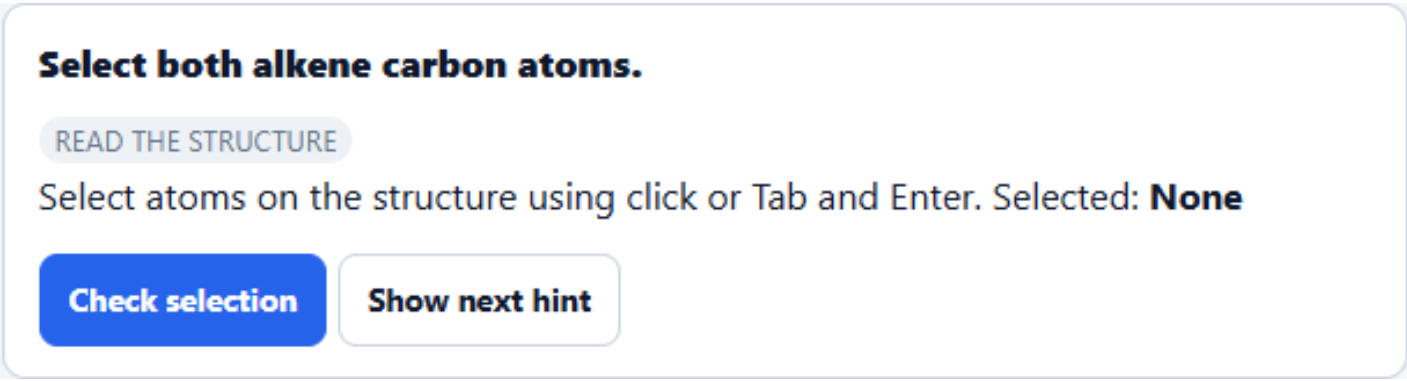
Screenshot: ez-ee-stage-1-molecule




## ez-ee:stage-1 | Select both alkene carbon atoms.

Answer type: atoms

Screenshot: ez-ee-stage-1-question



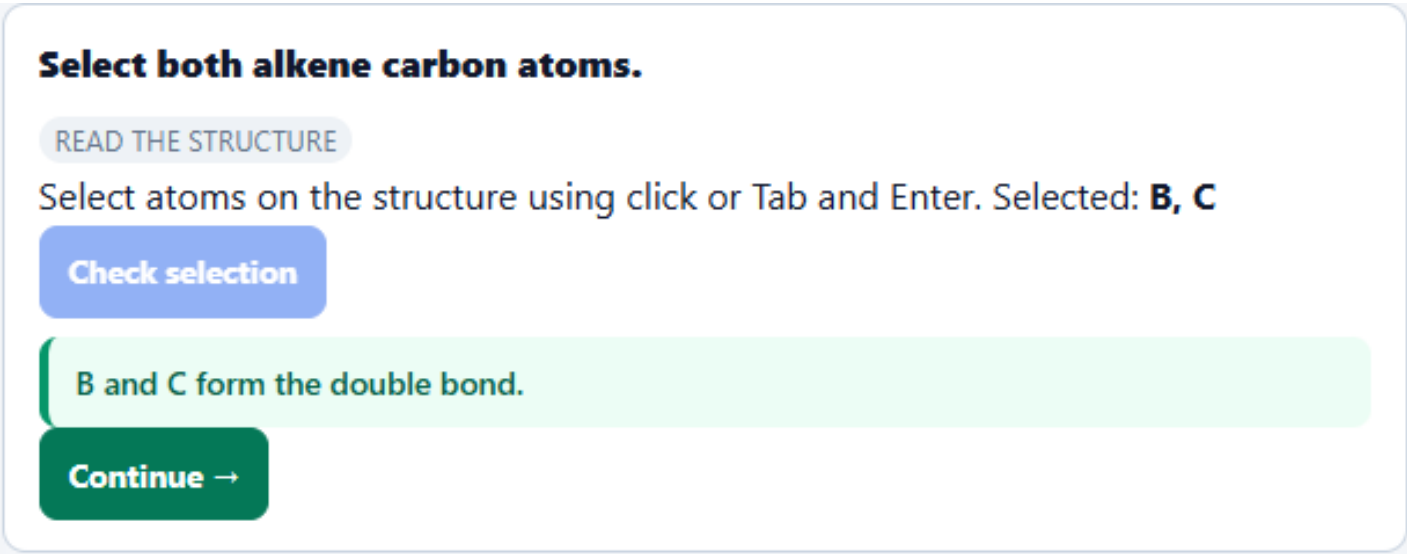

{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Look for two parallel bond lines.

Hint 2: Choose both endpoints of that bond.

Explanation: B and C form the double bond.

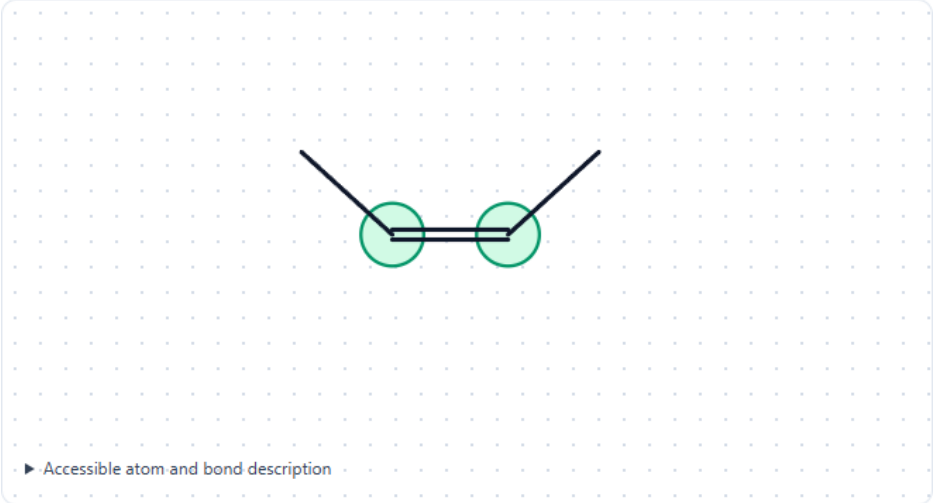
Screenshot: ez-ee-stage-1-feedback



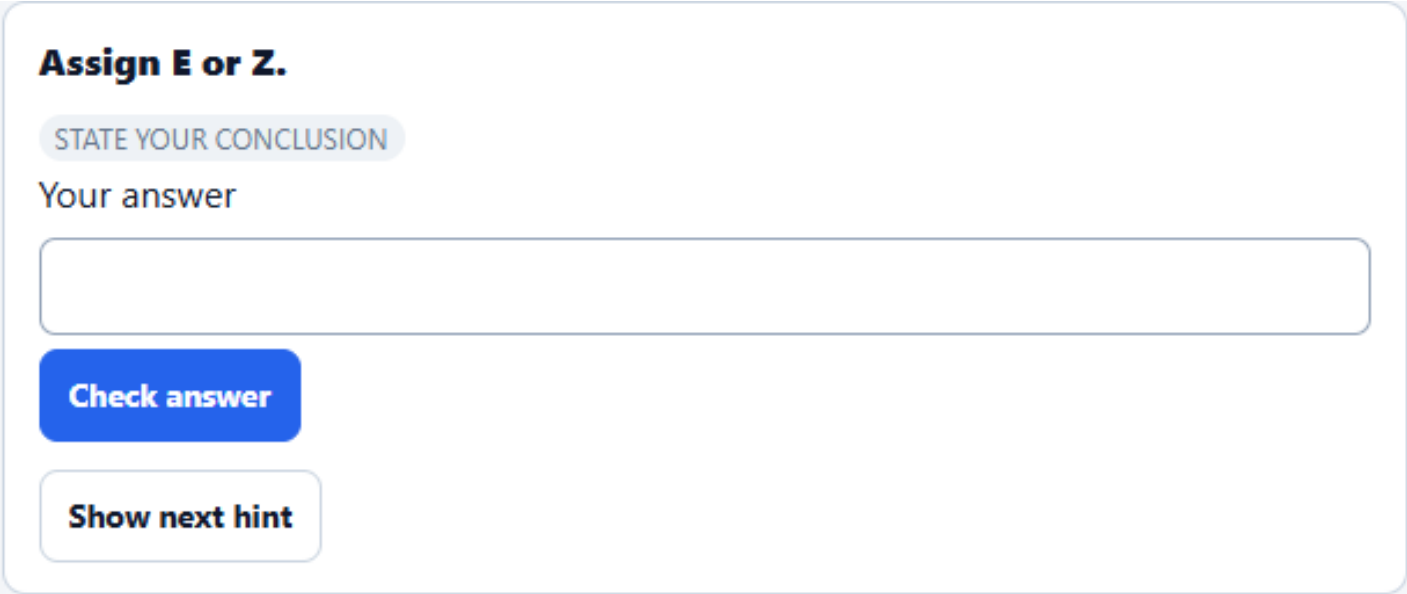

## ez-ee:stage-2 | Assign E or Z.

Answer type: text

Screenshot: ez-ee-stage-2-molecule



Screenshot: ez-ee-stage-2-question



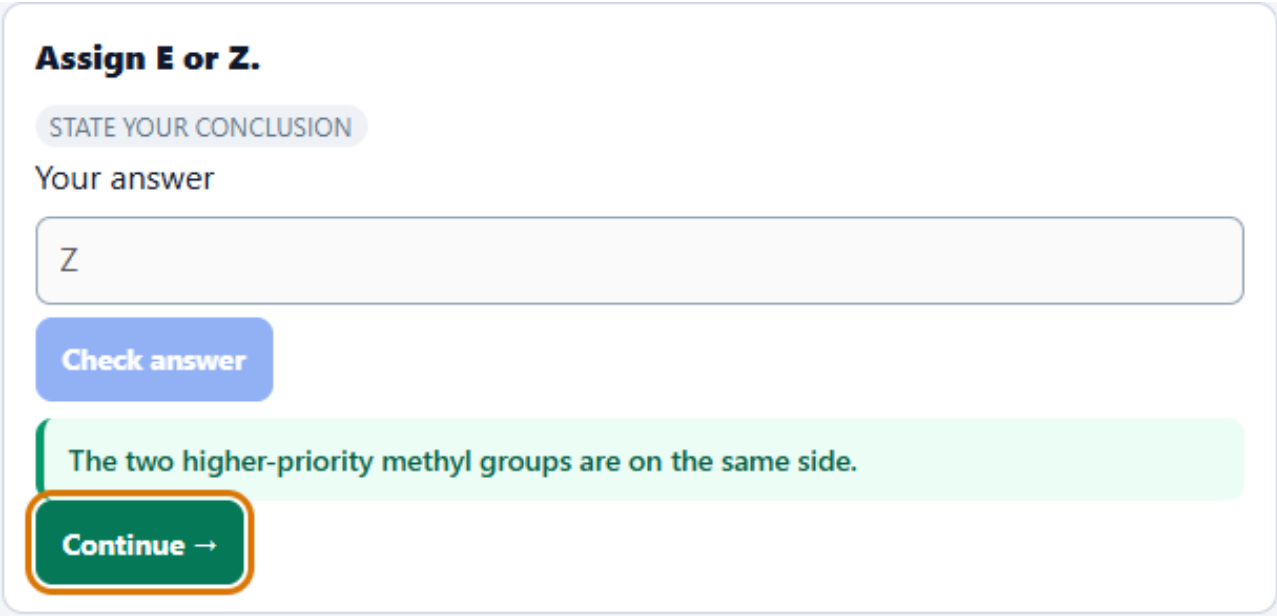

{"authorAnswer": ["Z", "(Z)"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rank methyl versus H on each end.

Hint 2: Same-side higher-priority groups determine the descriptor.

Explanation: The two higher-priority methyl groups are on the same side.

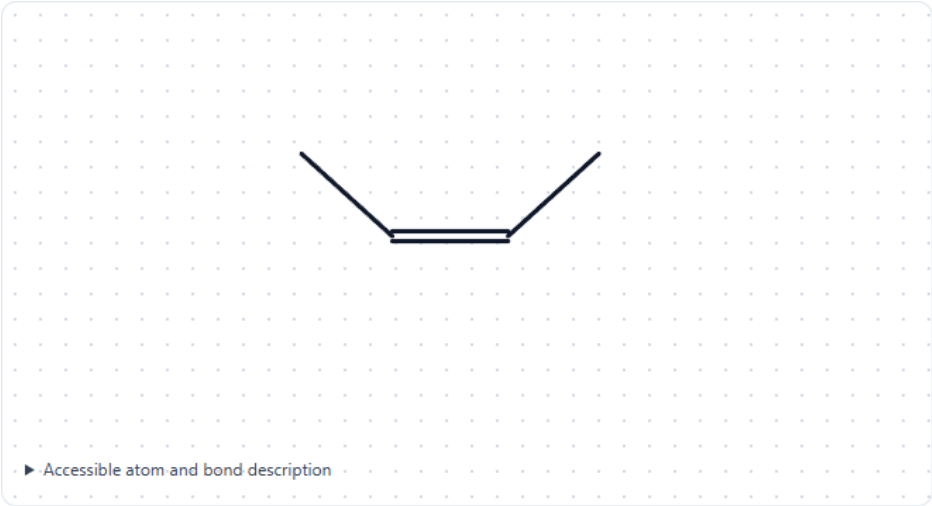
Screenshot: ez-ee-stage-2-feedback



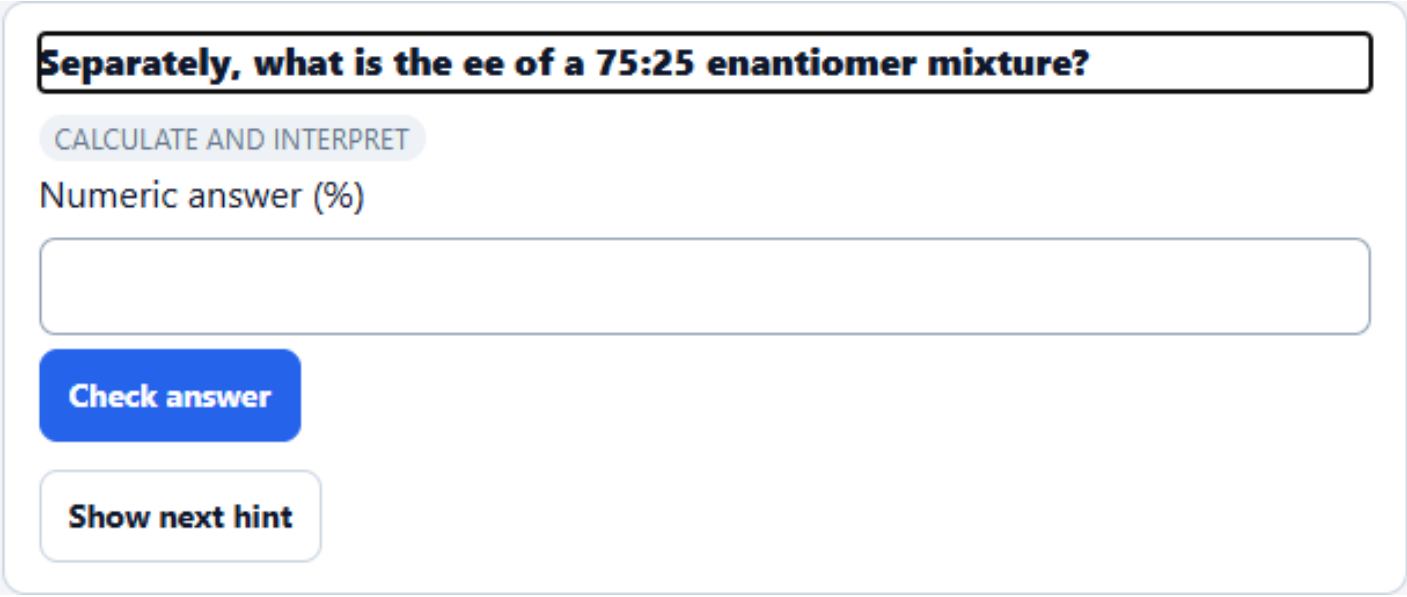

## ez-ee:stage-3 | Separately, what is the ee of a 75:25 enantiomer mixture?

Answer type: number | unit: %

Screenshot: ez-ee-stage-3-molecule




Screenshot: ez-ee-stage-3-question



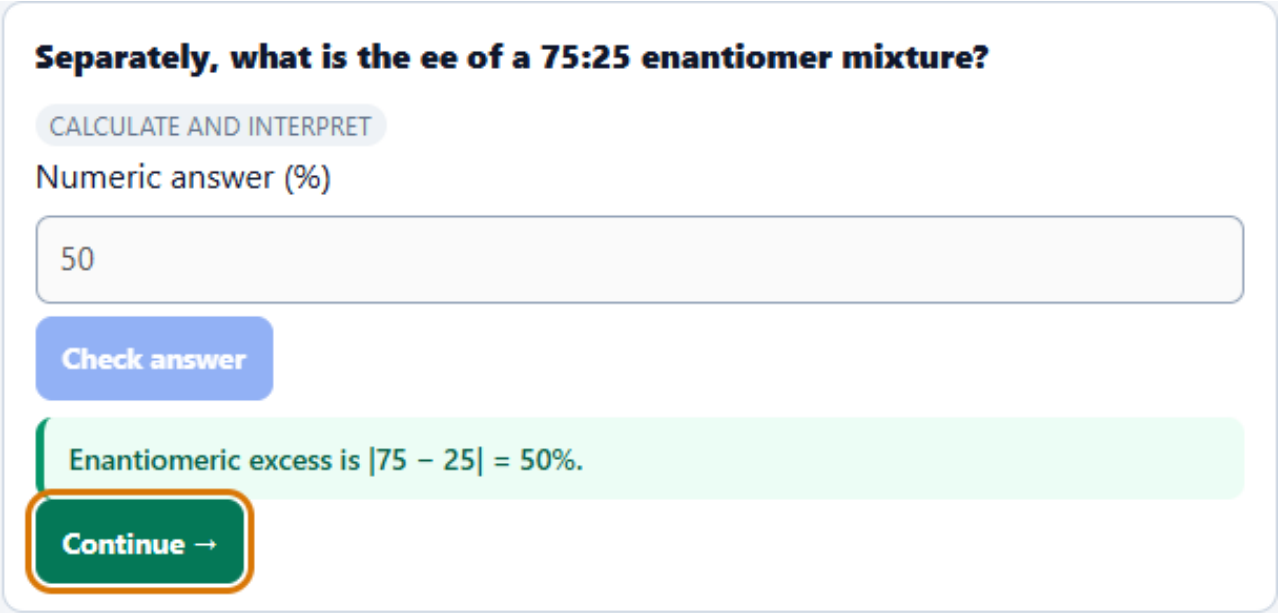

{"authorAnswer": 50, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Subtract the smaller percentage from the larger.

Hint 2: Do not divide the percentages.

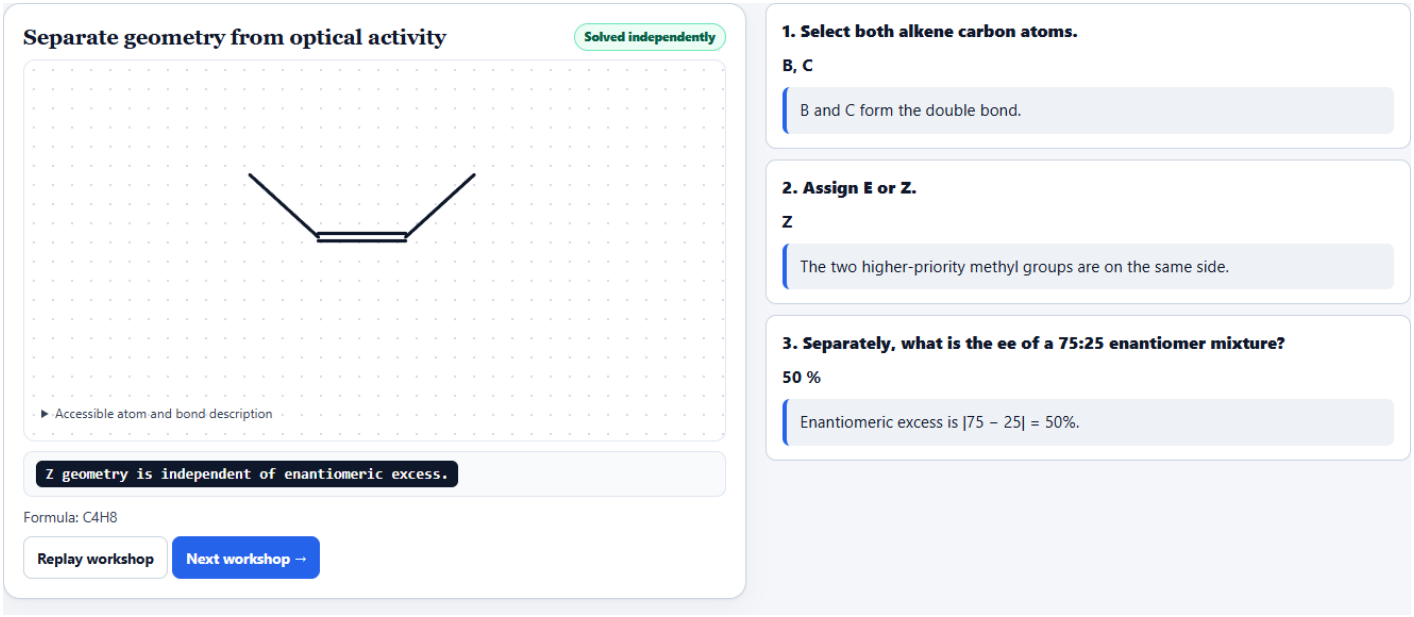
Explanation: Enantiomeric excess is |75 − 25| = 50%.

Screenshot: ez-ee-stage-3-feedback




Completion summary: Z geometry is independent of enantiomeric excess.

Screenshot: ez-ee-complete



# Molecular repository: author claims to verify

## ethanol

Author claims: ethanol | formula C2H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

## ether

Author claims: methoxymethane | formula C2H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=110.00; charge=0; explicit H override=none

1 (B): O; x=120.00, y=75.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=110.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

## isobutane

Author claims: 2-methylpropane | formula C4H10

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

## butane

Author claims: butane | formula C4H10

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1]]

## cyclohexane

Author claims: cyclohexane (connectivity diagram) | formula C6H12

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 0, 1]]

## butanol2

Author claims: butan-2-ol (connectivity; stereochemistry specified by projection) | formula C4H10O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

## butanediol

Author claims: butane-2,3-diol (stereochemistry unspecified) | formula C4H10O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=165.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1], [2, 5, 1]]

## e-butene

Author claims: (E)-but-2-ene | formula C4H8

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=240.00, y=150.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

## z-butene

Author claims: (Z)-but-2-ene | formula C4H8

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

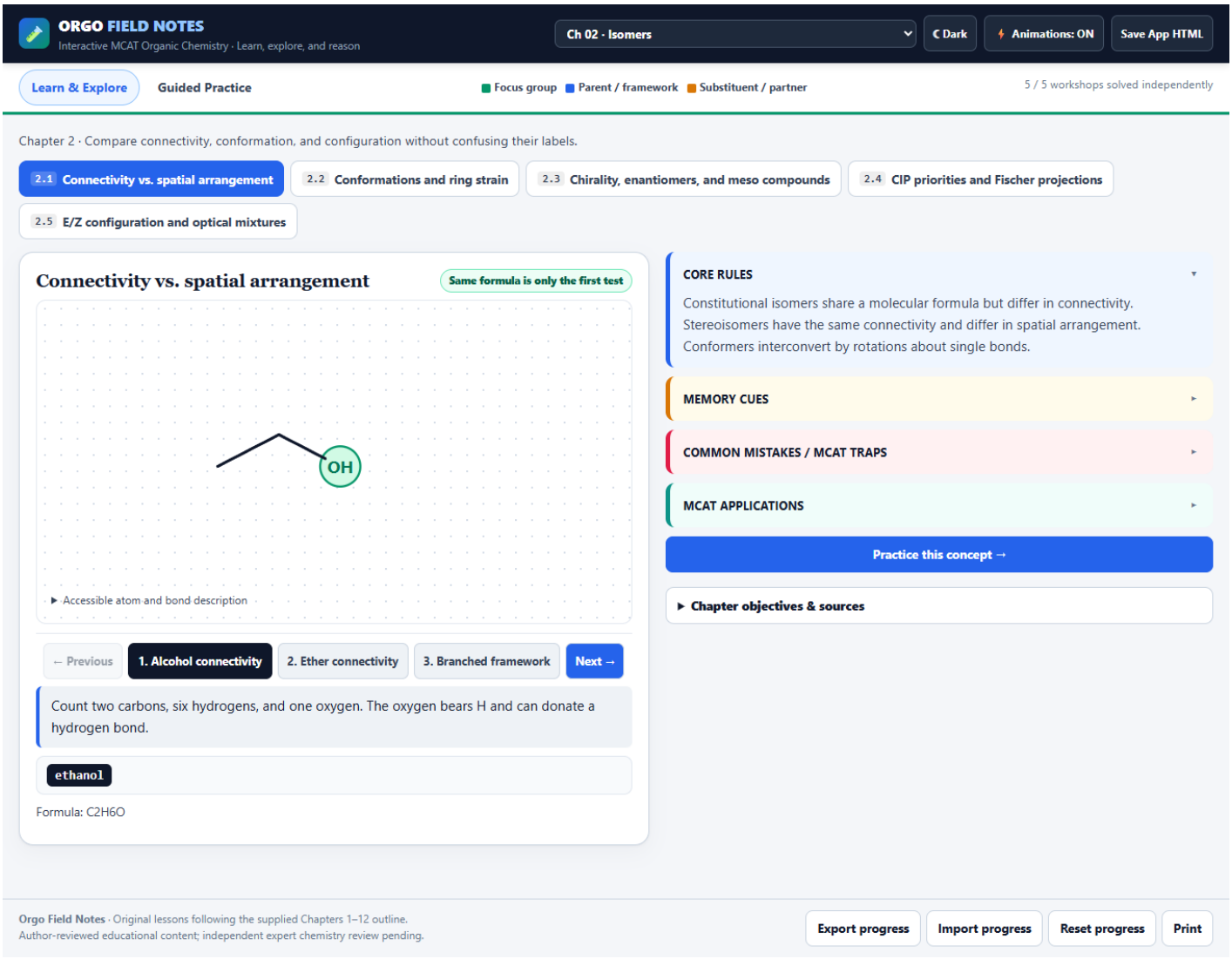
3 (D): C; x=240.00, y=50.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

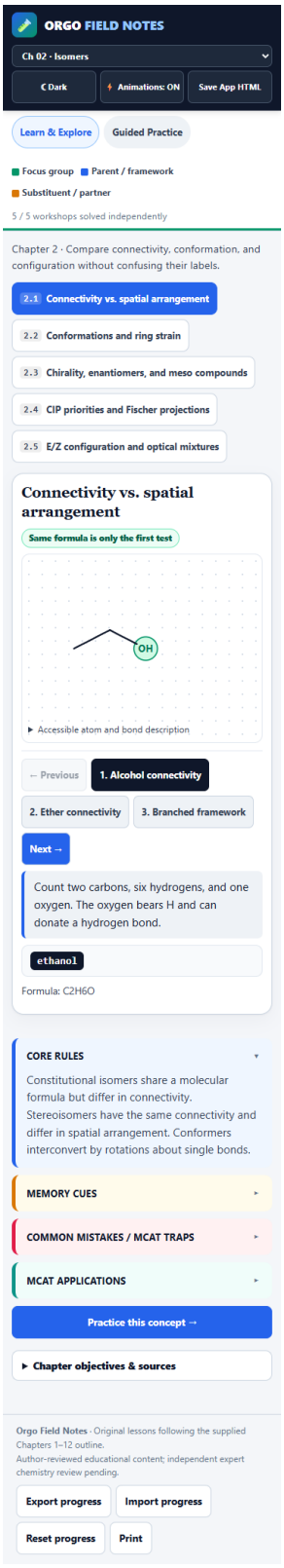
# Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

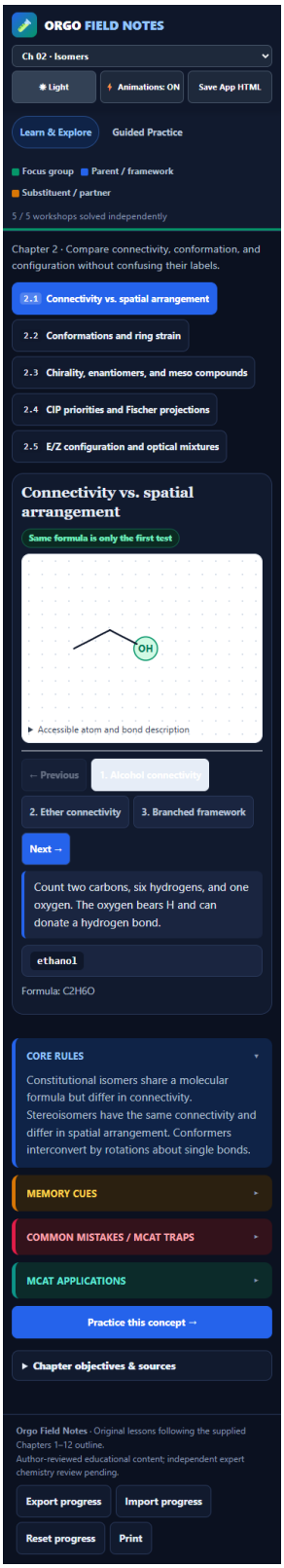
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark




# Execution evidence

72 screenshots and 16 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.